MIDDLESEX SCHOOL

MATHEMATICS DEPARTMENT

Mathematics Curriculum

Math 12 through Math 49

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Math 12 Curriculum

Intermediate Algebra | Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e

COURSE AT A GLANCE

Review Content

REVIEW

1.1 Use the Language of Algebra; 1.2 Integers; 1.3 Fractions; 1.4 Decimals; 1.5 Properties of Real Numbers

UNIT LEARNING OBJECTIVES

I can evaluate numerical expressions using the order of operations.

I can perform operations with signed integers and fractions.

I can simplify an algebraic expression by combining like terms.

I can use the distributive property to rewrite an expression.

I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

UNIT LEARNING OBJECTIVES

I can solve a linear equation and present equivalent steps clearly.

I can solve a linear equation containing fractions.

I can rearrange a formula to isolate a specified variable.

I can translate a verbal relationship into a linear equation.

I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

UNIT LEARNING OBJECTIVES

I can solve and graph a one-variable linear inequality.

I can solve a compound inequality involving AND.

I can solve a compound inequality involving OR.

I can express an inequality solution using interval notation.

I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

I can determine and interpret the slope of a line.

I can graph a line from an equation in a common form.

I can write an equation of a line from given information.

I can determine whether two lines are parallel, perpendicular, or neither.

I can evaluate a function using function notation.

I can determine whether a relation is a function using the vertical line test or input-output data.

I can determine domain and range from a graph or table.

I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

UNIT LEARNING OBJECTIVES

I can interpret the solution of a linear system as an intersection point.

I can solve a two-variable linear system by graphing.

I can solve a two-variable linear system by substitution.

I can solve a two-variable linear system by elimination.

I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

UNIT LEARNING OBJECTIVES

I can simplify expressions using integer exponent rules.

I can add and subtract polynomials.

I can multiply polynomials, including binomials.

I can divide a polynomial by a monomial.

I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

2.7 Solve Absolute Value Inequalities; 3.4 Graph Linear Inequalities in Two Variables; 6.1 Greatest Common Factor and Factor by Grouping; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

I can solve and graph an absolute-value inequality.

I can graph a system of linear inequalities and identify its solution region.

I can factor a quadratic trinomial.

I can identify a relationship as discrete or continuous.

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

Textbook coverage: 1.1 Use the Language of Algebra; 1.2 Integers; 1.3 Fractions; 1.4 Decimals; 1.5 Properties of Real Numbers

M12-REV-001

I can evaluate numerical expressions using the order of operations.

Textbook: 1.1 Use the Language of Algebra

SUPPORTING SKILLS

R Apply the order of operations to a numerical expression.
SK-NUM-ORDER-OPERATIONS · inherited from middle school

M12-REV-002

I can perform operations with signed integers and fractions.

Textbook: 1.2 Integers; 1.3 Fractions; 1.4 Decimals

SUPPORTING SKILLS

R Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school

R Add, subtract, multiply, and divide signed integers.
SK-NUM-INTEGER-OPERATE · inherited from middle school

M12-REV-003

I can simplify an algebraic expression by combining like terms.

Textbook: 1.5 Properties of Real Numbers

SUPPORTING SKILLS

- R Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- R Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school

M12-REV-004

I can use the distributive property to rewrite an expression.

Textbook: 1.5 Properties of Real Numbers

SUPPORTING SKILLS

- R Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school

M12-REV-005

I can determine the prime factorization of a whole number.

Textbook: 1.1 Use the Language of Algebra

SUPPORTING SKILLS

- R Express a whole number as a product of prime factors.
  SK-NUM-PRIME-FACTOR · inherited from middle school

Linear Equations and Applications REQUIRED

Textbook coverage: 2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

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M12-EQN-001

I can solve a linear equation and present equivalent steps clearly.

Textbook: 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

- I Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION
- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- I Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS
- A Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school

M12-EQN-002

I can solve a linear equation containing fractions.

Textbook: 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- I Clear fractions from an equation using a common denominator.
  SK-ALG-CLEAR-FRACTIONS
- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school

M12-EQN-003

I can rearrange a formula to isolate a specified variable.

Textbook: 2.3 Solve a Formula for a Specific Variable

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE

M12-EQN-004

I can translate a verbal relationship into a linear equation.

Textbook: 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION

M12-EQN-005

I can solve and interpret a number, mixture, or motion problem using a linear equation.

Textbook: 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

SUPPORTING SKILLS

- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

Inequalities and Absolute-Value Equations

REQUIRED

Textbook coverage: 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

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M12-INE-001

I can solve and graph a one-variable linear inequality.

Textbook: 2.5 Solve Linear Inequalities

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE
- I Reverse an inequality when multiplying or dividing by a negative value.
  SK-INE-REVERSE-SIGN

M12-INE-002

I can solve a compound inequality involving AND.

Textbook: 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Interpret AND as the intersection of two inequality solution sets.
  SK-INE-COMPOUND-AND
- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12
- A Reverse an inequality when multiplying or dividing by a negative value.
  SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-003

I can solve a compound inequality involving OR.

Textbook: 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Interpret OR as the union of two inequality solution sets.
  SK-INE-COMPOUND-OR
- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12
- A Reverse an inequality when multiplying or dividing by a negative value.
  SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-004

I can express an inequality solution using interval notation.

Textbook: 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12
- I Read inequality symbols and state the relationship each symbol expresses.
  SK-NOT-INEQUALITY-READ
- I Translate between inequality and interval notation.
  SK-NOT-INTERVAL

M12-INE-005

I can solve an absolute-value equation and check its solutions.

Textbook: 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

- I Rewrite an absolute-value equation as two cases.
  SK-ABS-SPLIT-EQUATION
- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

Lines and Introductory Functions REQUIRED

Textbook coverage: 3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions; 3.6 Graphs of Functions

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M12-LIN-001

I can determine and interpret the slope of a line.

Textbook: 3.2 Slope of a Line

SUPPORTING SKILLS

- I Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH
- I Calculate slope from two points.
  SK-LIN-SLOPE-POINTS
- I Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS

M12-LIN-002

I can graph a line from an equation in a common form.

Textbook: 3.1 Graph Linear Equations in Two Variables

SUPPORTING SKILLS

- I Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT
- I Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE
- I Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT

M12-LIN-003

I can write an equation of a line from given information.

Textbook: 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M12-LIN-004

I can determine whether two lines are parallel, perpendicular, or neither.

Textbook: 3.2 Slope of a Line; 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- I Recognize that distinct parallel lines have equal slopes.
  SK-LIN-PARALLEL-SLOPE
- I Determine the negative reciprocal of a nonzero slope.
  SK-LIN-PERPENDICULAR-SLOPE

M12-LIN-005

I can evaluate a function using function notation.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT
- I Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ
- I Recall and correctly use the terms relation, function, input, output, domain, and range.
  SK-FUNC-VOCABULARY

M12-LIN-006

I can determine whether a relation is a function using the vertical line test or input-output data.

Textbook: 3.5 Relations and Functions

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE
- A Recall and correctly use the terms relation, function, input, output, domain, and range.
  SK-FUNC-VOCABULARY · first introduced in Math 12

M12-LIN-008

I can determine domain and range from a graph or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- I Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH
- I Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE

M12-LIN-009

I can interpret domain and range in context.

Textbook: 3.6 Graphs of Functions

SUPPORTING SKILLS

- I Restrict domain to values meaningful in a context.
  SK-FUNC-CONTEXT-DOMAIN
- I Interpret the range of a function in a context.
  SK-FUNC-CONTEXT-RANGE
- I Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN

Systems of Linear Equations

REQUIRED

Textbook coverage: 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

M12-SYS-001

I can interpret the solution of a linear system as an intersection point.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- I — Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION

M12-SYS-002

I can solve a two-variable linear system by graphing.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A — Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- A — Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- I — Check an ordered pair in both equations of a system.
  SK-SYS-CHECK
- D — Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M12-SYS-003

I can solve a two-variable linear system by substitution.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A — Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- I — Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE

M12-SYS-004

I can solve a two-variable linear system by elimination.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A — Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- I — Add equations to eliminate a variable.
  SK-SYS-ELIMINATE

M12-SYS-005

I can create and solve a two-variable linear system for an application.

Textbook: 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

SUPPORTING SKILLS

- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- A Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- I Define two variables and write two equations for a context.
  SK-SYS-MODEL
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

Polynomial Operations and Basic Factoring

REQUIRED

Textbook coverage: 5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

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M12-POL-001

I can simplify expressions using integer exponent rules.

Textbook: 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- I Apply the power rule for exponents.
  SK-EXP-POWER
- I Apply the product rule for exponents.
  SK-EXP-PRODUCT
- I Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT

M12-POL-002

I can add and subtract polynomials.

Textbook: 5.1 Add and Subtract Polynomials

SUPPORTING SKILLS

- A Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school
- I Identify like polynomial terms by variable part and exponent.
  SK-POLY-LIKE-TERMS
- I Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY

M12-POL-003

I can multiply polynomials, including binomials.

Textbook: 5.3 Multiply Polynomials

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- I Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY
- I Multiply a monomial by a polynomial.
  SK-POLY-DISTRIBUTE
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-004

I can divide a polynomial by a monomial.

Textbook: 5.4 Dividing Polynomials

SUPPORTING SKILLS

- **A** Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- **I** Divide each term of a polynomial by a monomial.
  SK-POLY-DIVIDE-MONOMIAL
- **A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-005

I can factor a greatest common factor from a polynomial.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- **A** Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- **I** Factor the greatest common monomial factor from a polynomial.
  SK-FACTOR-GCF
- **I** Determine the numerical and variable greatest common factor of polynomial terms.
  SK-POLY-GCF
- **A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

Extension Content

EXTENSION

Textbook coverage: 2.7 Solve Absolute Value Inequalities; 3.4 Graph Linear Inequalities in Two Variables; 6.1 Greatest Common Factor and Factor by Grouping; 3.6 Graphs of Functions

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M12-EXT-001

I can solve and graph an absolute-value inequality.

Textbook: 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

- I — Rewrite an absolute-value inequality as a compound inequality.
  SK-ABS-SPLIT-INEQUALITY
- A — Interpret AND as the intersection of two inequality solution sets.
  SK-INE-COMPOUND-AND · first introduced in Math 12
- A — Interpret OR as the union of two inequality solution sets.
  SK-INE-COMPOUND-OR · first introduced in Math 12
- A — Represent an inequality solution on a number line.
  SK-INE-NUMBER-LINE · first introduced in Math 12

M12-EXT-002

I can graph a system of linear inequalities and identify its solution region.

Textbook: 3.4 Graph Linear Inequalities in Two Variables

SUPPORTING SKILLS

- I — Graph the boundary line of a linear inequality with the correct line type.
  SK-INE-GRAPH-BOUNDARY
- I — Identify the overlapping solution region of multiple linear inequalities.
  SK-INE-SYSTEM-REGION
- A — Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12

M12-EXT-003

I can factor a quadratic trinomial.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- I — Verify a factorization by multiplication.
  SK-FACTOR-CHECK
- I — Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL
- I — Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC

M12-EXT-004

I can identify a relationship as discrete or continuous.

Textbook: 3.6 Graphs of Functions

SUPPORTING SKILLS

- I — Distinguish discrete inputs from values varying continuously over an interval.
  SK-FUNC-DISCRETE-CONTINUOUS

Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

COURSE AT A GLANCE

Review Content

REVIEW

3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 5.2 Properties of Exponents and Scientific Notation

UNIT LEARNING OBJECTIVES

I can determine and interpret the slope of a line.

I can graph a line from an equation and write an equation from given information.

I can apply a linear function to a context.

I can identify the solution of a linear system as an intersection point.

I can solve a simple two-variable linear system using an appropriate method.

I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

3.5 Relations and Functions; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

I can evaluate a function from a formula, graph, or table.

I can determine whether a relation is a function.

I can determine domain and range from a graph or table.

I can solve for an input that produces a specified output.

I can interpret inputs, outputs, domain, and range in context.

I can calculate and interpret average rate of change.

I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

UNIT LEARNING OBJECTIVES

I can add and subtract polynomials.

I can multiply polynomials.

I can factor out a greatest common factor.

I can factor a quadratic trinomial.

I can factor a difference of squares and other required special cases.

I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

UNIT LEARNING OBJECTIVES

I can state excluded values for a rational expression.

I can simplify a rational expression while preserving its restrictions.

I can multiply and divide rational expressions.

I can add and subtract rational expressions using a common denominator.

I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

8.3 Simplify Rational Exponents; 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

UNIT LEARNING OBJECTIVES

I can simplify expressions containing negative exponents.

I can rewrite expressions between radical and rational-exponent forms.

I can simplify radical expressions.

I can apply exponent rules to expressions with integer and rational exponents.

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations

UNIT LEARNING OBJECTIVES

I can identify key features of a quadratic function from its graph or equation.

I can connect standard, factored, and vertex forms to features of a parabola.

I can solve a quadratic equation by factoring.

I can solve a quadratic equation using the square-root property.

I can solve a quadratic equation by completing the square.

I can solve a quadratic equation using the quadratic formula.

I can choose an efficient method for solving a quadratic equation.

I can graph a quadratic function using its vertex, intercepts, and symmetry.

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

9.3 Solve Quadratic Equations Using the Quadratic Formula

UNIT LEARNING OBJECTIVES

I can perform basic arithmetic with complex numbers.

I can express the nonreal solutions of a quadratic equation using complex numbers.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

Textbook coverage: 3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 5.2 Properties of Exponents and Scientific Notation

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M21-REV-001

I can determine and interpret the slope of a line.

Textbook: 3.2 Slope of a Line

SUPPORTING SKILLS

- R Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- R Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12
- R Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-REV-002

I can graph a line from an equation and write an equation from given information.

Textbook: 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- R Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- R Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- R Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M21-REV-003

I can apply a linear function to a context.

Textbook: 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- I Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET
- R Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-REV-004

I can identify the solution of a linear system as an intersection point.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- R Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations

SUPPORTING SKILLS

- R Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- R Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- R Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- R Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

Textbook: 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- R Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- R Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- R Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12

Functions and Function Interpretation

REQUIRED

Textbook coverage: 3.5 Relations and Functions; 3.6 Graphs of Functions

M21-FUN-001

I can evaluate a function from a formula, graph, or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ · first introduced in Math 12

M21-FUN-002

I can determine whether a relation is a function.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M21-FUN-003

I can determine domain and range from a graph or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M21-FUN-004

I can solve for an input that produces a specified output.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT

M21-FUN-005

I can interpret inputs, outputs, domain, and range in context.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D Restrict domain to values meaningful in a context.
  SK-FUNC-CONTEXT-DOMAIN · first introduced in Math 12
- D Interpret the range of a function in a context.
  SK-FUNC-CONTEXT-RANGE · first introduced in Math 12
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12

M21-FUN-006

I can calculate and interpret average rate of change.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Calculate average rate of change over an interval.
  SK-FUNC-AROC
- I Interpret average rate of change with contextual units.
  SK-FUNC-AROC-INTERPRET
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-FUN-007

I can compare functions represented by formulas, graphs, or tables.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP
- I Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH
- I Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH

Polynomials and Factoring

REQUIRED

Textbook coverage: 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

M21-POL-001

I can add and subtract polynomials.

Textbook: 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school
- D Identify like polynomial terms by variable part and exponent.
  SK-POLY-LIKE-TERMS · first introduced in Math 12
- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-002

I can multiply polynomials.

Textbook: 5.3 Multiply Polynomials; 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- D Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12
- D Multiply a monomial by a polynomial.
  SK-POLY-DISTRIBUTE · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-003

I can factor out a greatest common factor.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- R Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- D Factor the greatest common monomial factor from a polynomial.
  SK-FACTOR-GCF · first introduced in Math 12
- A Determine the numerical and variable greatest common factor of polynomial terms.
  SK-POLY-GCF · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-004

I can factor a quadratic trinomial.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials

SUPPORTING SKILLS

- A Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- R Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- R Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-005

I can factor a difference of squares and other required special cases.

Textbook: 6.3 Factor Special Products

SUPPORTING SKILLS

- A Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- I Factor a difference of two squares.
  SK-FACTOR-DIFF-SQUARES
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-006

I can use factoring to determine the zeros of a polynomial.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Factor a difference of two squares.
  SK-FACTOR-DIFF-SQUARES · first introduced in Math 21
- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- I Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

Rational Expressions and Equations

REQUIRED

Textbook coverage: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

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M21-RAT-001

I can state excluded values for a rational expression.

Textbook: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

- I Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS
- I Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
  SK-RAT-VOCABULARY

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

Textbook: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- I Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS
- I Factor numerators and denominators completely.
  SK-RAT-FACTOR
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21
- A Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
  SK-RAT-VOCABULARY · first introduced in Math 21

M21-RAT-003

I can multiply and divide rational expressions.

Textbook: 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- A Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- A Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

Textbook: 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- I Combine rational expressions over a common denominator.
  SK-RAT-COMBINE
- I Determine the least common denominator of rational expressions.
  SK-RAT-LCD

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

Textbook: 7.5 Solve Rational Equations

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Clear denominators from a rational equation.
  SK-RAT-CLEAR-DENOMINATORS
- I Check candidate solutions against denominator restrictions.
  SK-RAT-EXTRANEOUS
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

Exponents and Radicals

REQUIRED

Textbook coverage: 8.3 Simplify Rational Exponents; 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

M21-EXP-001

I can simplify expressions containing negative exponents.

Textbook: 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I — Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE
- A — Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- A — Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12

M21-EXP-002

I can rewrite expressions between radical and rational-exponent forms.

Textbook: 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I — Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL
- I — Convert between radical and rational-exponent notation.
  SK-RAD-CONVERT
- I — Read radical and rational-exponent notation and identify the index, radicand, base, numerator, and denominator.
  SK-RAD-NOTATION-READ

M21-EXP-003

I can simplify radical expressions.

Textbook: 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

- A — Express a whole number as a product of prime factors.
  SK-NUM-PRIME-FACTOR · inherited from middle school
- I — Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT

M21-EXP-004

I can apply exponent rules to expressions with integer and rational exponents.

Textbook: 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

- A Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE · first introduced in Math 21
- D Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- D Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- D Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21
- I Apply the zero-exponent rule.
  SK-EXP-ZERO

M21-EXP-005

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT
- I Check a candidate solution for extraneous results.
  SK-RAD-CHECK
- I Isolate a radical expression before solving.
  SK-RAD-ISOLATE

Quadratic Functions and Equations REQUIRED

Textbook coverage: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations

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M21-QUAD-001

I can identify key features of a quadratic function from its graph or equation.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- I Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY
- I Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS
- I Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX
- I Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
  SK-QUAD-VOCABULARY

M21-QUAD-002

I can connect standard, factored, and vertex forms to features of a parabola.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- I Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT
- D Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- A Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-003

I can solve a quadratic equation by factoring.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- I Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS

M21-QUAD-004

I can solve a quadratic equation using the square-root property.

Textbook: 9.1 Solve Quadratic Equations Using the Square Root Property

SUPPORTING SKILLS

- I Apply the square-root property with both signs.
  SK-QUAD-SQUARE-ROOT
- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-005

I can solve a quadratic equation by completing the square.

Textbook: 9.2 Solve Quadratic Equations by Completing the Square

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE

M21-QUAD-006

I can solve a quadratic equation using the quadratic formula.

Textbook: 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- I Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT
- I Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA
- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-007

I can choose an efficient method for solving a quadratic equation.

Textbook: 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form

SUPPORTING SKILLS

- A Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- A Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA · first introduced in Math 21
- I Select an efficient quadratic-solving method from equation structure.
  SK-QUAD-METHOD-SELECT
- A Apply the square-root property with both signs.
  SK-QUAD-SQUARE-ROOT · first introduced in Math 21

M21-QUAD-008

I can graph a quadratic function using its vertex, intercepts, and symmetry.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-009

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Textbook: 9.5 Solve Applications of Quadratic Equations

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- I Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

Textbook coverage: 9.3 Solve Quadratic Equations Using the Quadratic Formula

M21-EXT-001

I can perform basic arithmetic with complex numbers.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Simplify integer powers of the imaginary unit.
  SK-CPLX-I-POWERS
- I Add, subtract, and multiply complex numbers.
  SK-CPLX-OPERATE

M21-EXT-002

I can express the nonreal solutions of a quadratic equation using complex numbers.

Textbook: 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- A Simplify integer powers of the imaginary unit.
  SK-CPLX-I-POWERS · first introduced in Math 21
- I Express a quadratic solution with a negative discriminant in complex form.
  SK-CPLX-QUAD-SOLUTION
- A Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT · first introduced in Math 21

Math 22 Curriculum

Geometry | Middlesex School Mathematics | Resource: Math 22 Geometry Packet

COURSE AT A GLANCE

Review Content

REVIEW

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can solve a linear equation arising from a geometric relationship.

I can solve a proportion and interpret the resulting scale factor.

I can simplify an exact square root and approximate it when needed.

I can calculate distance, midpoint, and slope on the coordinate plane.

I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED

1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

UNIT LEARNING OBJECTIVES

I can name and represent points, lines, planes, segments, and rays using correct notation.

I can identify collinear, coplanar, intersecting, and concurrent figures.

I can calculate segment lengths using betweenness and the segment-addition postulate.

I can determine a midpoint or use a segment bisector relationship.

I can classify and measure angles.

I can calculate angle measures using angle addition and angle bisectors.

I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

I can distinguish a definition, postulate, theorem, converse, and counterexample.

I can determine angle relationships formed by parallel lines and a transversal.

I can use angle relationships to determine whether lines are parallel.

I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

I can construct a logical geometric argument from stated facts and accepted results.

Triangle Congruence and Proof

REQUIRED

2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

UNIT LEARNING OBJECTIVES

I can classify triangles by sides and angles.

I can use triangle angle-sum and exterior-angle relationships.

I can write a congruence statement with corresponding vertices in the correct order.

I can determine whether triangles are congruent by SAS.

I can determine whether triangles are congruent by SSS.

I can determine whether right triangles are congruent by HL.

I can determine whether triangles are congruent by ASA or AAS.

I can apply isosceles and equilateral triangle theorems and their converses.

I can use corresponding parts of congruent triangles after proving congruence.

I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED

3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

UNIT LEARNING OBJECTIVES

I can calculate interior or exterior angle measures of a polygon.

I can calculate perimeter and area of common polygons.

I can apply properties of parallelograms to determine unknown measures.

I can determine whether a quadrilateral is a parallelogram.

I can apply properties of rectangles, rhombi, and squares.

Quadrilaterals, Perimeter, and Area (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can classify a quadrilateral from its stated or marked properties.

I can apply trapezoid and trapezoid-midsegment relationships.

I can apply properties of kites.

I can calculate the perimeter or area of a composite polygon.

I can justify a quadrilateral conclusion using definitions and theorems.

Similarity and Proportional Reasoning

REQUIRED

4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

UNIT LEARNING OBJECTIVES

I can determine image lengths and coordinates under a dilation.

I can identify the scale factor of a dilation or similarity relationship.

I can write a similarity statement with corresponding vertices in the correct order.

I can use proportional corresponding sides to solve similar-polygon problems.

I can relate similarity scale factor to perimeter and area scale factors.

I can prove triangles similar by AA, SSS, or SAS.

I can apply triangle proportionality theorems and their converses.

I can solve and justify an indirect-measurement or other similarity application.

Right Triangles

REQUIRED

5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

UNIT LEARNING OBJECTIVES

I can apply the Pythagorean theorem to determine a missing side.

I can use the converse of the Pythagorean theorem to classify a triangle.

I can apply triangle-inequality relationships.

I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

I can apply right-triangle similarity relationships.

I can use geometric-mean relationships created by an altitude to the hypotenuse.

I can select sine, cosine, or tangent for a right-triangle problem.

I can determine a missing side or angle of a right triangle.

I can solve a contextual right-triangle problem.

I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED

6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

UNIT LEARNING OBJECTIVES

I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

I can calculate circumference and arc length.

I can calculate the area of a circle, sector, or segment.

I can determine angle and arc measures involving central and inscribed angles.

I can apply relationships among chords, arcs, and distances from the center.

I can apply tangent-radius and tangent-segment relationships.

I can determine angle measures formed by chords, secants, or tangents.

I can determine segment lengths formed by intersecting chords.

I can determine segment lengths formed by secants or tangents.

I can solve a multi-step circle problem and justify each relationship used.

SKILL PROGRESSION — I Introduce · R Reinforce · D Deepen · A Apply

Review Content

REVIEW

Resource coverage: No mapped textbook section

M22-REV-001

I can solve a linear equation arising from a geometric relationship.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A — Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- R — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A — Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

M22-REV-002

I can solve a proportion and interpret the resulting scale factor.

Resource: No mapped textbook section

SUPPORTING SKILLS

- R — Use cross multiplication when two ratios form a proportion.
  SK-ALG-CROSS-MULTIPLY · inherited from middle school
- I — Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR

M22-REV-003

I can simplify an exact square root and approximate it when needed.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Distinguish exact values from measured or rounded approximations.
  SK-GEO-EXACT-APPROX
- R Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M22-REV-004

I can calculate distance, midpoint, and slope on the coordinate plane.

Resource: No mapped textbook section

SUPPORTING SKILLS

- R Calculate horizontal or vertical distance from the difference of coordinates.
  SK-COORD-DISTANCE-AXIS · inherited horizontal/vertical coordinate distance
- I Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN
- R Calculate the midpoint of a segment from endpoint coordinates.
  SK-COORD-MIDPOINT · inherited from earlier mathematics
- R Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M22-REV-005

I can substitute measurements into a formula with consistent units.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS
- A Apply the order of operations to a numerical expression.
  SK-NUM-ORDER-OPERATIONS · inherited from middle school

Geometry Basics REQUIRED

Resource coverage: 1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

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M22-BAS-001

I can name and represent points, lines, planes, segments, and rays using correct notation.

Resource: 1.1 Points, Lines, Planes

SUPPORTING SKILLS

- I Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK
- I Recall definitions of points, lines, planes, segments, rays, angles, and intersections.
  SK-GEO-FOUNDATIONAL-DEFINITIONS
- I Read and write geometric notation correctly.
  SK-GEO-NOTATION

M22-BAS-002

I can identify collinear, coplanar, intersecting, and concurrent figures.

Resource: 1.1 Points, Lines, Planes

SUPPORTING SKILLS

- I Distinguish stated or marked facts from visual appearance.
  SK-GEO-DIAGRAM-INFER
- A Read and write geometric notation correctly.
  SK-GEO-NOTATION · first introduced in Math 22

M22-BAS-003

I can calculate segment lengths using betweenness and the segment-addition postulate.

Resource: 1.2 Segments, Measurement, Addition, Midpoints, Bisectors

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I Translate segment addition into an equation.
  SK-GEO-SEGMENT-ADDITION

M22-BAS-004

I can determine a midpoint or use a segment bisector relationship.

Resource: 1.2 Segments, Measurement, Addition, Midpoints, Bisectors

SUPPORTING SKILLS

- I Use a bisector to create equal measures.
  SK-GEO-BISECTOR
- I Use a midpoint to create equal segment lengths.
  SK-GEO-MIDPOINT
- A Translate segment addition into an equation.
  SK-GEO-SEGMENT-ADDITION · first introduced in Math 22

M22-BAS-005

I can classify and measure angles.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs

SUPPORTING SKILLS

- I Identify complementary, supplementary, vertical, and linear-pair relationships.
  SK-ANG-PAIR-IDENTIFY
- I Measure or calculate an angle accurately.
  SK-GEO-ANGLE-MEASURE

M22-BAS-006

I can calculate angle measures using angle addition and angle bisectors.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Translate angle addition into an equation.
  SK-GEO-ANGLE-ADDITION
- A Use a bisector to create equal measures.
  SK-GEO-BISECTOR · first introduced in Math 22

M22-BAS-007

I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- D Identify complementary, supplementary, vertical, and linear-pair relationships.
  SK-ANG-PAIR-IDENTIFY · first introduced in Math 22

M22-BAS-008

I can distinguish a definition, postulate, theorem, converse, and counterexample.

Resource: 1.1 Points, Lines, Planes; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- I Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE
- I Disprove a universal statement with a counterexample.
  SK-PROOF-COUNTEREXAMPLE
- I Apply a geometric definition as justification.
  SK-PROOF-DEFINITION
- I Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT

M22-BAS-009

I can determine angle relationships formed by parallel lines and a transversal.

Resource: 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- I — Use parallel-line angle relationships.
  SK-ANG-PARALLEL-APPLY
- I — Identify corresponding, alternate, and same-side angle pairs.
  SK-ANG-TRANSVERSAL-IDENTIFY

M22-BAS-010

I can use angle relationships to determine whether lines are parallel.

Resource: 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- I — Use angle relationships to prove lines parallel.
  SK-ANG-PARALLEL-CONVERSE
- A — Identify corresponding, alternate, and same-side angle pairs.
  SK-ANG-TRANSVERSAL-IDENTIFY · first introduced in Math 22

M22-BAS-011

I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

Resource: 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A — Use parallel-line angle relationships.
  SK-ANG-PARALLEL-APPLY · first introduced in Math 22
- I — Use right-angle relationships created by perpendicular lines.
  SK-LIN-PERPENDICULAR-ANGLE

M22-BAS-012

I can construct a logical geometric argument from stated facts and accepted results.

Resource: 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- I — Translate givens into diagram markings and statements.
  SK-PROOF-GIVEN
- I — Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE
- A — Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Triangle Congruence and Proof

REQUIRED

Resource coverage: 2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

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M22-CON-001

I can classify triangles by sides and angles.

Resource: 2.1 Triangles, Sides, and Angles

SUPPORTING SKILLS

- A — Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK · first introduced in Math 22
- I — Classify a triangle by side lengths and angle measures.
  SK-TRI-CLASSIFY
- I — Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.
  SK-TRI-CONGRUENCE-VOCAB

M22-CON-002

I can use triangle angle-sum and exterior-angle relationships.

Resource: 2.1 Triangles, Sides, and Angles

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I — Apply the triangle angle-sum theorem.
  SK-TRI-ANGLE-SUM
- I — Apply the triangle exterior-angle theorem.
  SK-TRI-EXTERIOR-ANGLE

M22-CON-003

I can write a congruence statement with corresponding vertices in the correct order.

Resource: 2.2 Congruent Polygons; SAS

SUPPORTING SKILLS

- I — Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE
- I — Write congruence statements in corresponding order.
  SK-PROOF-CONGRUENCE-ORDER
- A — Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.
  SK-TRI-CONGRUENCE-VOCAB · first introduced in Math 22

M22-CON-004

I can determine whether triangles are congruent by SAS.

Resource: 2.2 Congruent Polygons; SAS

SUPPORTING SKILLS

- A — Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- D — Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- I — Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT

M22-CON-005

I can determine whether triangles are congruent by SSS.

Resource: 2.3 SSS and HL

SUPPORTING SKILLS

- A Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- D Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-006

I can determine whether right triangles are congruent by HL.

Resource: 2.3 SSS and HL

SUPPORTING SKILLS

- A Use right-angle relationships created by perpendicular lines.
  SK-LIN-PERPENDICULAR-ANGLE · first introduced in Math 22
- D Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-007

I can determine whether triangles are congruent by ASA or AAS.

Resource: 2.5 ASA and AAS

SUPPORTING SKILLS

- A Apply the triangle angle-sum theorem.
  SK-TRI-ANGLE-SUM · first introduced in Math 22
- D Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-008

I can apply isosceles and equilateral triangle theorems and their converses.

Resource: 2.4 Isosceles and Equilateral Triangles

SUPPORTING SKILLS

- D Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- I Apply isosceles-triangle theorems and converses.
  SK-TRI-ISOSCELES

M22-CON-009

I can use corresponding parts of congruent triangles after proving congruence.

Resource: 2.2 Congruent Polygons; SAS; 2.5 ASA and AAS

SUPPORTING SKILLS

- A Write congruence statements in corresponding order.
  SK-PROOF-CONGRUENCE-ORDER · first introduced in Math 22
- I Use corresponding parts only after triangle congruence is established.
  SK-PROOF-CPCTC

M22-CON-010

I can write a structured proof involving triangle congruence.

Resource: 2.3 SSS and HL; 2.4 Isosceles and Equilateral Triangles; 2.5 ASA and AAS

SUPPORTING SKILLS

- A Use corresponding parts only after triangle congruence is established.
  SK-PROOF-CPCTC · first introduced in Math 22
- A Translate givens into diagram markings and statements.
  SK-PROOF-GIVEN · first introduced in Math 22
- D Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE · first introduced in Math 22
- A Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

Quadrilaterals, Perimeter, and Area

REQUIRED

Resource coverage: 3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

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M22-QUAD-001

I can calculate interior or exterior angle measures of a polygon.

Resource: 3.1 Polygon Angles, Perimeter, and Area

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Calculate polygon interior or exterior angle sums.
  SK-POLY-ANGLE-SUM

M22-QUAD-002

I can calculate perimeter and area of common polygons.

Resource: 3.1 Polygon Angles, Perimeter, and Area

SUPPORTING SKILLS

- I Add boundary lengths without omitting or duplicating sides.
  SK-GEO-PERIMETER
- D Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22
- A Apply the order of operations to a numerical expression.
  SK-NUM-ORDER-OPERATIONS · inherited from middle school

M22-QUAD-003

I can apply properties of parallelograms to determine unknown measures.

Resource: 3.2 Parallelograms

SUPPORTING SKILLS

- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I Apply parallelogram properties and tests.
  SK-QUAD-PARALLELOGRAM

M22-QUAD-004

I can determine whether a quadrilateral is a parallelogram.

Resource: 3.2 Parallelograms

SUPPORTING SKILLS

- A Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- D Apply parallelogram properties and tests.
  SK-QUAD-PARALLELOGRAM · first introduced in Math 22

M22-QUAD-005

I can apply properties of rectangles, rhombi, and squares.

Resource: 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- I — Use the inclusive hierarchy of quadrilateral classifications.
  SK-QUAD-HIERARCHY
- I — Distinguish and apply rectangle, rhombus, and square properties.
  SK-QUAD-RECT-RHOM-SQUARE

M22-QUAD-006

I can classify a quadrilateral from its stated or marked properties.

Resource: 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- A — Distinguish stated or marked facts from visual appearance.
  SK-GEO-DIAGRAM-INFER · first introduced in Math 22
- I — Recall the definitions and defining properties of the major quadrilateral families.
  SK-QUAD-DEFINITIONS
- D — Use the inclusive hierarchy of quadrilateral classifications.
  SK-QUAD-HIERARCHY · first introduced in Math 22

M22-QUAD-007

I can apply trapezoid and trapezoid-midsegment relationships.

Resource: 3.4 Trapezoids

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I — Apply trapezoid and midsegment relationships.
  SK-QUAD-TRAPEZOID

M22-QUAD-008

I can apply properties of kites.

Resource: 3.5 Kites and Composite Polygons

SUPPORTING SKILLS

- I — Apply kite properties.
  SK-QUAD-KITE

M22-QUAD-009

I can calculate the perimeter or area of a composite polygon.

Resource: 3.5 Kites and Composite Polygons

SUPPORTING SKILLS

- I — Decompose a composite region into familiar shapes.
  SK-GEO-AREA-DECOMPOSE
- D — Add boundary lengths without omitting or duplicating sides.
  SK-GEO-PERIMETER · first introduced in Math 22
- A — Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22

M22-QUAD-010

I can justify a quadrilateral conclusion using definitions and theorems.

Resource: 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- D Apply a geometric definition as justification.
  SK-PROOF-DEFINITION · first introduced in Math 22
- A Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE · first introduced in Math 22
- D Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Similarity and Proportional Reasoning REQUIRED

Resource coverage: 4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

M22-SIM-001

I can determine image lengths and coordinates under a dilation.

Resource: 4.1 Dilations and Similarity

SUPPORTING SKILLS

- A Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- D Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22

M22-SIM-002

I can identify the scale factor of a dilation or similarity relationship.

Resource: 4.1 Dilations and Similarity

SUPPORTING SKILLS

- A Use cross multiplication when two ratios form a proportion.
  SK-ALG-CROSS-MULTIPLY · inherited from middle school
- D Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22
- I Recall and correctly use the terms similarity, dilation, scale factor, and corresponding parts.
  SK-SIM-VOCAB

M22-SIM-003

I can write a similarity statement with corresponding vertices in the correct order.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- D Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- I Write similarity statements in corresponding order.
  SK-PROOF-SIMILARITY-ORDER

M22-SIM-004

I can use proportional corresponding sides to solve similar-polygon problems.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- A Use cross multiplication when two ratios form a proportion.
  SK-ALG-CROSS-MULTIPLY · inherited from middle school
- A Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- I Set up proportions using corresponding sides.
  SK-SIM-PROPORTION

M22-SIM-005

I can relate similarity scale factor to perimeter and area scale factors.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- I Relate length scale factor to perimeter and area factors.
  SK-SIM-PERIMETER-AREA
- A Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22

M22-SIM-006

I can prove triangles similar by AA, SSS, or SAS.

Resource: 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS

SUPPORTING SKILLS

- A Write similarity statements in corresponding order.
  SK-PROOF-SIMILARITY-ORDER · first introduced in Math 22
- D Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE · first introduced in Math 22
- I Recognize AA triangle similarity.
  SK-SIM-AA
- I Test SSS or SAS triangle similarity.
  SK-SIM-SSS-SAS

M22-SIM-007

I can apply triangle proportionality theorems and their converses.

Resource: 4.5 Proportionality Theorems

SUPPORTING SKILLS

- A Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- I Apply triangle proportionality relationships.
  SK-SIM-TRI-PROPORTIONALITY

M22-SIM-008

I can solve and justify an indirect-measurement or other similarity application.

Resource: 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS

SUPPORTING SKILLS

- A Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- D Set up proportions using corresponding sides.
  SK-SIM-PROPORTION · first introduced in Math 22
- A Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22

Right Triangles

REQUIRED

Resource coverage: 5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

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M22-RGT-001

I can apply the Pythagorean theorem to determine a missing side.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- I Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN

M22-RGT-002

I can use the converse of the Pythagorean theorem to classify a triangle.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- A Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- D Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22

M22-RGT-003

I can apply triangle-inequality relationships.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- A Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- A Classify a triangle by side lengths and angle measures.
  SK-TRI-CLASSIFY · first introduced in Math 22

M22-RGT-004

I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

Resource: 5.2 Special Right Triangles

SUPPORTING SKILLS

- A Distinguish exact values from measured or rounded approximations.
  SK-GEO-EXACT-APPROX · first introduced in Math 22
- R Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- I Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS

M22-RGT-005

I can apply right-triangle similarity relationships.

Resource: 5.3 HL Similarity and Right-Triangle Similarity

SUPPORTING SKILLS

- A Recognize AA triangle similarity.
  SK-SIM-AA · first introduced in Math 22
- D Set up proportions using corresponding sides.
  SK-SIM-PROPORTION · first introduced in Math 22

M22-RGT-006

I can use geometric-mean relationships created by an altitude to the hypotenuse.

Resource: 5.3 HL Similarity and Right-Triangle Similarity

SUPPORTING SKILLS

- A Set up proportions using corresponding sides.
  SK-SIM-PROPORTION · first introduced in Math 22
- I Apply geometric-mean relationships in a right triangle split by an altitude.
  SK-TRI-GEOMETRIC-MEAN

M22-RGT-007

I can select sine, cosine, or tangent for a right-triangle problem.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK · first introduced in Math 22
- I Recall and correctly use the terms leg, hypotenuse, opposite side, adjacent side, and trigonometric ratio.
  SK-TRI-RIGHT-VOCAB
- I Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO
- I Label sides relative to a selected angle.
  SK-TRIG-TRIANGLE-LABEL

M22-RGT-008

I can determine a missing side or angle of a right triangle.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- I Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE
- I Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE

M22-RGT-009

I can solve a contextual right-triangle problem.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- A Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- A Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22

M22-RGT-010

I can justify a right-triangle solution using an appropriate theorem or ratio.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- D Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- A Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22
- A Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS · first introduced in Math 22
- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22

Circles

REQUIRED

Resource coverage: 6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

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M22-CIR-001

I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

Resource: 6.1 Circle Vocabulary, Circumference, Arc Length

SUPPORTING SKILLS

- A Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB · first introduced in Math 22
- A Read and write geometric notation correctly.
  SK-GEO-NOTATION · first introduced in Math 22

M22-CIR-002

I can calculate circumference and arc length.

Resource: 6.1 Circle Vocabulary, Circumference, Arc Length

SUPPORTING SKILLS

- I Relate a central angle to a fraction of a circle.
  SK-CIR-ARC-FRACTION
- I Calculate arc length from radius and central angle.
  SK-CIR-ARC-LENGTH
- I Calculate circumference from radius or diameter.
  SK-CIR-CIRCUMFERENCE
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22

M22-CIR-003

I can calculate the area of a circle, sector, or segment.

Resource: 6.2 Circle, Sector, and Segment Area

SUPPORTING SKILLS

- I Calculate the area of a circle.
  SK-CIR-AREA
- I Calculate sector area from radius and central angle.
  SK-CIR-SECTOR-AREA
- I Calculate segment area by subtracting a triangle from a sector.
  SK-CIR-SEGMENT-AREA
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22

M22-CIR-004

I can determine angle and arc measures involving central and inscribed angles.

Resource: 6.3 Angle, Arc, and Chord Relationships

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Relate a central angle to a fraction of a circle.
  SK-CIR-ARC-FRACTION · first introduced in Math 22
- I Relate an inscribed angle to its intercepted arc.
  SK-CIR-INSCRIBED-ANGLE

M22-CIR-005

I can apply relationships among chords, arcs, and distances from the center.

Resource: 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships

SUPPORTING SKILLS

- I Apply equal-chord, equal-arc, and center-distance relationships.
  SK-CIR-CHORD-RELATION
- I Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB

M22-CIR-006

I can apply tangent-radius and tangent-segment relationships.

Resource: 6.4 Tangent, Secant, and Chord Relationships

SUPPORTING SKILLS

- I Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT
- I Use the perpendicular relationship between a radius and tangent.
  SK-CIR-TANGENT-RADIUS

M22-CIR-007

I can determine angle measures formed by chords, secants, or tangents.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- I Select the correct angle formula for chords, secants, or tangents.
  SK-CIR-ANGLE-RELATION
- A Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB · first introduced in Math 22

M22-CIR-008

I can determine segment lengths formed by intersecting chords.

Resource: 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT

M22-CIR-009

I can determine segment lengths formed by secants or tangents.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT · first introduced in Math 22

M22-CIR-010

I can solve a multi-step circle problem and justify each relationship used.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A Select the correct angle formula for chords, secants, or tangents.
  SK-CIR-ANGLE-RELATION · first introduced in Math 22
- A Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT · first introduced in Math 22
- A Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB · first introduced in Math 22
- D Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Math 31 Curriculum

Advanced Algebra | Middlesex School Mathematics | Textbook: OpenStax College Algebra 2e (Jay Abramson, 2021)

COURSE AT A GLANCE

Review Content

REVIEW

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can calculate a percentage, percent increase, or percent decrease.

I can simplify an expression using exponent properties.

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED

3.1 Functions and Function Notation; 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions; 3.2 Domain and Range; 4.1 Linear Functions

UNIT LEARNING OBJECTIVES

I can interpret function notation as a statement about input and output.

I can determine whether a graph or table represents a function.

I can evaluate a function or solve for an input using a graph, table, or formula.

I can identify independent and dependent variables in a relationship.

I can calculate average rate of change from two points.

I can interpret interval notation and identify increasing or decreasing intervals.

I can determine total change from an average rate of change over an interval.

I can determine and interpret the units of an average rate of change.

I can identify a linear function from a table, graph, or formula.

I can interpret the slope and intercept of a linear model.

I can determine the intersection of two linear relationships graphically or algebraically.

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.5 Transformation of Functions; 3.4 Composition of Functions; 3.7 Inverse Functions; 3.3 Rates of Change and Behavior of Graphs

UNIT LEARNING OBJECTIVES

I can determine whether a relation is a function.

I can evaluate a function from a formula, graph, or table.

I can determine domain and range from a formula, graph, or table.

I can determine algebraic domain restrictions caused by denominators and even roots.

I can graph a piecewise function from its formula.

I can write a piecewise formula that represents a graph.

I can graph a transformed function using shifts, stretches, compressions, and reflections.

I can describe how a transformation affects domain, range, or asymptotes.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for a composition of two functions.

I can find an inverse function algebraically.

I can evaluate and interpret an inverse from a graph, formula, or table.

I can determine and interpret the units of an inverse function.

I can explain how the domain and range of a function relate to those of its inverse.

I can determine the range of a function by analyzing the domain of its inverse.

I can verify algebraically that two functions are inverses.

I can interpret the meaning and units of a composition.

I can classify a graph or table as concave up or concave down using changes in average rate of change.

Function Behavior and Structure (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine whether a function is invertible using the horizontal line test.

I can graph an inverse function as a reflection across the line `y = x`.

Quadratic Functions and Equations

REQUIRED

5.1 Quadratic Functions; 2.5 Quadratic Equations

UNIT LEARNING OBJECTIVES

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

I can connect standard, factored, and vertex forms to features of a quadratic graph.

I can rewrite a quadratic function in a form that reveals a requested feature.

I can solve a quadratic equation by factoring or using the quadratic formula.

I can graph a quadratic function using its algebraic structure and key features.

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

I can create and interpret a quadratic trajectory model.

Exponential Functions and Models

REQUIRED

6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

UNIT LEARNING OBJECTIVES

I can distinguish linear change from exponential change using differences, ratios, or percent change.

I can identify initial value and growth or decay factor in an exponential model.

I can write an exponential function from a table, graph, or context.

I can evaluate and interpret an exponential model.

I can calculate growth under periodic compounding.

I can calculate growth under continuous compounding.

I can determine an exponential formula from two points or other sufficient data.

I can determine domain, range, intercepts, and asymptotes of an exponential function.

I can compare the growth rates of two exponential functions.

I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

6.3 Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models; 6.4 Graphs of Logarithmic Functions

UNIT LEARNING OBJECTIVES

I can convert between logarithmic and exponential forms.

I can evaluate a logarithm using its exponential meaning.

I can expand or condense an expression using product, quotient, and power properties.

I can solve an exponential equation using logarithms.

I can solve a logarithmic equation using exponential form and check domain restrictions.

I can calculate and interpret doubling time or half-life.

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

I can explain how exponential and logarithmic functions undo one another.

I can solve a multi-step exponential or logarithmic equation.

I can determine an intersection of exponential functions graphically or algebraically.

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

3.6 Absolute Value Functions

UNIT LEARNING OBJECTIVES

I can analyze an absolute-value function using transformations.

I can determine even or odd symmetry from a graph or formula.

I can interpret a contextual logarithmic scale.

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

Textbook coverage: No mapped textbook section

M31-REV-001

I can calculate a percentage, percent increase, or percent decrease.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R Calculate a percentage of a quantity or determine what percent one quantity is of another.
  SK-NUM-PERCENT-CALCULATE · inherited from middle school
- R Calculate and interpret percent increase or decrease.
  SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-REV-002

I can simplify an expression using exponent properties.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE · first introduced in Math 21
- R Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- R Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- R Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- R Apply the zero-exponent rule.
  SK-EXP-ZERO · first introduced in Math 21

M31-REV-003

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M31-REV-004

I can expand a product of two binomials.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- R Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12

Linear Functions and Function Notation REQUIRED

Textbook coverage: 3.1 Functions and Function Notation; 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions; 3.2 Domain and Range; 4.1 Linear Functions

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M31-LIN-001

I can interpret function notation as a statement about input and output.

Textbook: 3.1 Functions and Function Notation

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ · first introduced in Math 12
- I Use function notation to express and solve an input-output question.
  SK-FUNC-NOTATION-SOLVE

M31-LIN-002

I can determine whether a graph or table represents a function.

Textbook: 3.1 Functions and Function Notation

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-LIN-003

I can evaluate a function or solve for an input using a graph, table, or formula.

Textbook: 3.1 Functions and Function Notation

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Use function notation to express and solve an input-output question.
  SK-FUNC-NOTATION-SOLVE · first introduced in Math 31
- D Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-LIN-004

I can identify independent and dependent variables in a relationship.

Textbook: 3.1 Functions and Function Notation

SUPPORTING SKILLS

- I Identify independent and dependent variables.
  SK-FUNC-INDEPENDENT-DEPENDENT
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12

M31-LIN-005

I can calculate average rate of change from two points.

Textbook: 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

- D Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M31-LIN-006

I can interpret interval notation and identify increasing or decreasing intervals.

Textbook: 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- I Describe increasing or decreasing behavior over intervals.
  SK-FUNC-INTERVAL-BEHAVIOR
- D Translate between inequality and interval notation.
  SK-NOT-INTERVAL · first introduced in Math 12

M31-LIN-007

I can determine total change from an average rate of change over an interval.

Textbook: 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

- A Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- I Use average rate of change and interval length to calculate total change.
  SK-FUNC-AROC-TOTAL-CHANGE
- A Add, subtract, multiply, and divide signed integers.
  SK-NUM-INTEGER-OPERATE · inherited from middle school

M31-LIN-008

I can determine and interpret the units of an average rate of change.

Textbook: 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

- I Determine rate units from input- and output-axis units.
  SK-FUNC-AROC-UNITS
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-LIN-009

I can identify a linear function from a table, graph, or formula.

Textbook: 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

- I Use constant differences or ratios to distinguish linear and exponential patterns.
  SK-EXP-DIFFERENCE-RATIO
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-LIN-010

I can interpret the slope and intercept of a linear model.

Textbook: 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

- A Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-LIN-011

I can determine the intersection of two linear relationships graphically or algebraically.

Textbook: 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- A Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- A Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M31-LIN-012

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Textbook: 4.1 Linear Functions

SUPPORTING SKILLS

- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- A Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- D Recognize that distinct parallel lines have equal slopes.
  SK-LIN-PARALLEL-SLOPE · first introduced in Math 12
- D Determine the negative reciprocal of a nonzero slope.
  SK-LIN-PERPENDICULAR-SLOPE · first introduced in Math 12

Function Behavior and Structure REQUIRED

Textbook coverage: 3.1 Functions and Function Notation; 3.2 Domain and Range; 3.5 Transformation of Functions; 3.4 Composition of Functions; 3.7 Inverse Functions; 3.3 Rates of Change and Behavior of Graphs

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M31-FUN-001

I can determine whether a relation is a function.

Textbook: 3.1 Functions and Function Notation

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-FUN-002

I can evaluate a function from a formula, graph, or table.

Textbook: 3.1 Functions and Function Notation

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-FUN-003

I can determine domain and range from a formula, graph, or table.

Textbook: 3.2 Domain and Range

SUPPORTING SKILLS

- I Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA
- D Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M31-FUN-004

I can determine algebraic domain restrictions caused by denominators and even roots.

Textbook: 3.2 Domain and Range

SUPPORTING SKILLS

- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

M31-FUN-005

I can graph a piecewise function from its formula.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Describe increasing or decreasing behavior over intervals.
  SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I Select the correct rule of a piecewise function for a given input.
  SK-FUNC-PIECEWISE-EVALUATE
- I Graph each rule on its stated domain with correct endpoints.
  SK-FUNC-PIECEWISE-GRAPH

M31-FUN-006

I can write a piecewise formula that represents a graph.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Describe increasing or decreasing behavior over intervals.
  SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I Translate a segmented graph into a piecewise formula.
  SK-FUNC-PIECEWISE-WRITE
- A Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12

M31-FUN-007

I can graph a transformed function using shifts, stretches, compressions, and reflections.

Textbook: 3.5 Transformation of Functions

SUPPORTING SKILLS

- I Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL
- I Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE
- I Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-FUN-008

I can describe how a transformation affects domain, range, or asymptotes.

Textbook: 3.5 Transformation of Functions

SUPPORTING SKILLS

- I — Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE
- D — Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D — Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- D — Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M31-FUN-009

I can evaluate a composition from formulas, graphs, or tables.

Textbook: 3.4 Composition of Functions

SUPPORTING SKILLS

- I — Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA
- I — Follow intermediate values to compose functions from tables.
  SK-FUNC-COMPOSE-TABLE
- A — Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I — Read and write notation for function composition and inverse functions.
  SK-FUNC-STRUCTURE-NOTATION

M31-FUN-010

I can write a formula for a composition of two functions.

Textbook: 3.4 Composition of Functions

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D — Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31

M31-FUN-011

I can find an inverse function algebraically.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- A — Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I — Exchange variables and solve to obtain an inverse formula.
  SK-FUNC-INVERSE-ALGEBRA
- I — Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP
- A — Read and write notation for function composition and inverse functions.
  SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M31-FUN-012

I can evaluate and interpret an inverse from a graph, formula, or table.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Exchange variables and solve to obtain an inverse formula.
  SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-013

I can determine and interpret the units of an inverse function.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- I Reverse input and output units when interpreting an inverse.
  SK-FUNC-INVERSE-UNITS
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-FUN-014

I can explain how the domain and range of a function relate to those of its inverse.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- I Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE
- A Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-015

I can determine the range of a function by analyzing the domain of its inverse.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- A Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31

M31-FUN-016

I can verify algebraically that two functions are inverses.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY

M31-FUN-017

I can interpret the meaning and units of a composition.

Textbook: 3.4 Composition of Functions

SUPPORTING SKILLS

- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Follow intermediate values to compose functions from tables.
  SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-FUN-018

I can classify a graph or table as concave up or concave down using changes in average rate of change.

Textbook: 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

- A Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- I Use successive average rates of change to classify concavity.
  SK-FUNC-CONCAVITY-AROC
- D Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-FUN-019

I can determine whether a function is invertible using the horizontal line test.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- I Apply the horizontal line test for invertibility.
  SK-FUNC-INVERSE-HLT
- I Test whether a function is one-to-one on a domain.
  SK-FUNC-ONE-TO-ONE

M31-FUN-020

I can graph an inverse function as a reflection across the line `y = x`.

Textbook: 3.7 Inverse Functions

SUPPORTING SKILLS

- A Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

Quadratic Functions and Equations REQUIRED

Textbook coverage: 5.1 Quadratic Functions; 2.5 Quadratic Equations

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M31-QUAD-001

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

Textbook: 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Use successive average rates of change to classify concavity.
  SK-FUNC-CONCAVITY-AROC · first introduced in Math 31
- A Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- R Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
  SK-QUAD-VOCABULARY · first introduced in Math 21

M31-QUAD-002

I can connect standard, factored, and vertex forms to features of a quadratic graph.

Textbook: 5.1 Quadratic Functions

SUPPORTING SKILLS

- D Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- A Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-QUAD-003

I can rewrite a quadratic function in a form that reveals a requested feature.

Textbook: 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- D Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-004

I can solve a quadratic equation by factoring or using the quadratic formula.

Textbook: 2.5 Quadratic Equations; 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT · first introduced in Math 21
- A Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA · first introduced in Math 21
- D Select an efficient quadratic-solving method from equation structure.
  SK-QUAD-METHOD-SELECT · first introduced in Math 21

M31-QUAD-005

I can graph a quadratic function using its algebraic structure and key features.

Textbook: 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-QUAD-006

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

Textbook: 5.1 Quadratic Functions

SUPPORTING SKILLS

- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT
- A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-007

I can create and interpret a quadratic trajectory model.

Textbook: 5.1 Quadratic Functions

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Exponential Functions and Models

REQUIRED

Textbook coverage: 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

M31-EXP-001

I can distinguish linear change from exponential change using differences, ratios, or percent change.

Textbook: 6.1 Exponential Functions

SUPPORTING SKILLS

- D Use constant differences or ratios to distinguish linear and exponential patterns.
  SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31
- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- A Calculate and interpret percent increase or decrease.
  SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-EXP-002

I can identify initial value and growth or decay factor in an exponential model.

Textbook: 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

- I Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT
- I Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR
- I Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
  SK-EXP-VOCABULARY
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M31-EXP-003

I can write an exponential function from a table, graph, or context.

Textbook: 6.1 Exponential Functions

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- I Determine an exponential model from tabular values.
  SK-EXP-MODEL-TABLE
- I Solve for an exponential model using two points.
  SK-EXP-MODEL-TWO-POINTS
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-004

I can evaluate and interpret an exponential model.

Textbook: 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-EXP-005

I can calculate growth under periodic compounding.

Textbook: 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- I Substitute correctly into a periodic-compounding model.
  SK-EXP-COMPOUND-PERIODIC
- A Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A Calculate a percentage of a quantity or determine what percent one quantity is of another.
  SK-NUM-PERCENT-CALCULATE · inherited from middle school

M31-EXP-006

I can calculate growth under continuous compounding.

Textbook: 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- I Substitute correctly into a continuous-compounding model.
  SK-EXP-COMPOUND-CONTINUOUS
- A Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31

M31-EXP-007

I can determine an exponential formula from two points or other sufficient data.

Textbook: 6.1 Exponential Functions

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Solve for an exponential model using two points.
  SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31
- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21

M31-EXP-008

I can determine domain, range, intercepts, and asymptotes of an exponential function.

Textbook: 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

- I Identify horizontal asymptotic behavior of an exponential graph.
  SK-EXP-ASYMPTOTE
- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-009

I can compare the growth rates of two exponential functions.

Textbook: 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

- I Compare exponential growth rates expressed in different forms.
  SK-EXP-COMPARE-GROWTH
- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21

M31-EXP-010

I can determine the percent growth or decay represented by a continuous exponential model.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Convert between equivalent discrete- and continuous-growth forms.
  SK-EXP-DISCRETE-CONTINUOUS
- D Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A Calculate and interpret percent increase or decrease.
  SK-NUM-PERCENT-CHANGE · inherited from middle school

Logarithmic Functions and Models REQUIRED

Textbook coverage: 6.3 Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models; 6.4 Graphs of Logarithmic Functions

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M31-LOG-001

I can convert between logarithmic and exponential forms.

Textbook: 6.3 Logarithmic Functions

SUPPORTING SKILLS

- I Convert between logarithmic and exponential forms.
  SK-LOG-CONVERT
- I Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION
- I Read logarithmic notation and identify the base, argument, and corresponding exponential statement.
  SK-LOG-NOTATION-READ

M31-LOG-002

I can evaluate a logarithm using its exponential meaning.

Textbook: 6.3 Logarithmic Functions

SUPPORTING SKILLS

- I Rewrite exponential expressions using a common base when possible.
  SK-EXP-REWRITE-BASE
- D Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-003

I can expand or condense an expression using product, quotient, and power properties.

Textbook: 6.5 Logarithmic Properties

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Apply the logarithm power property in either direction.
  SK-LOG-POWER
- I Apply the logarithm product property in either direction.
  SK-LOG-PRODUCT
- I Apply the logarithm quotient property in either direction.
  SK-LOG-QUOTIENT

M31-LOG-004

I can solve an exponential equation using logarithms.

Textbook: 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- A Apply the logarithm power property in either direction.
  SK-LOG-POWER · first introduced in Math 31
- I Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP

M31-LOG-005

I can solve a logarithmic equation using exponential form and check domain restrictions.

Textbook: 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Convert between logarithmic and exponential forms.
  SK-LOG-CONVERT · first introduced in Math 31
- I Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK
- I Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG

M31-LOG-006

I can calculate and interpret doubling time or half-life.

Textbook: 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- I Translate a doubling-time or half-life question into an exponential equation.
  SK-LOG-DOUBLING-HALF
- A Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M31-LOG-007

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

Textbook: 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- I Identify vertical asymptotic behavior of a logarithmic graph.
  SK-LOG-ASYMPTOTE
- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-LOG-008

I can explain how exponential and logarithmic functions undo one another.

Textbook: 6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

- A Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- A Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-009

I can solve a multi-step exponential or logarithmic equation.

Textbook: 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A Apply the logarithm power property in either direction.
  SK-LOG-POWER · first introduced in Math 31
- A Apply the logarithm product property in either direction.
  SK-LOG-PRODUCT · first introduced in Math 31
- A Apply the logarithm quotient property in either direction.
  SK-LOG-QUOTIENT · first introduced in Math 31
- D Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- D Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31

M31-LOG-010

I can determine an intersection of exponential functions graphically or algebraically.

Textbook: 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

- A Compare exponential growth rates expressed in different forms.
  SK-EXP-COMPARE-GROWTH · first introduced in Math 31
- D Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- A Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- A Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M31-LOG-011

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- D Convert between equivalent discrete- and continuous-growth forms.
  SK-EXP-DISCRETE-CONTINUOUS · first introduced in Math 31
- A Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31

M31-LOG-012

I can solve and interpret an exponential or logarithmic modeling problem.

Textbook: 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- A Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

Textbook coverage: 3.6 Absolute Value Functions

M31-EXT-001

I can analyze an absolute-value function using transformations.

Textbook: 3.6 Absolute Value Functions

SUPPORTING SKILLS

- A Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- A Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXT-002

I can determine even or odd symmetry from a graph or formula.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY

M31-EXT-003

I can interpret a contextual logarithmic scale.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31
- I Interpret equal intervals on a logarithmic scale as equal multiplicative changes.
  SK-LOG-SCALE-INTERPRET
- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

Math 32 Curriculum

Precalculus: Trigonometry | Middlesex School Mathematics | Resource: Math 32 Workbook

COURSE AT A GLANCE

Trigonometric Foundations

REQUIRED

pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

UNIT LEARNING OBJECTIVES

I can use the Pythagorean theorem to determine an unknown side of a right triangle.

I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

I can convert angle measures between degrees and radians.

I can determine coterminal and reference angles for a given angle.

I can calculate arc length from a central angle and radius.

I can determine exact coordinates on the unit circle.

I can evaluate all six trigonometric functions exactly for common angles.

I can solve right triangles using trigonometric ratios.

I can apply right-triangle trigonometry to contextual problems.

I can evaluate trigonometric functions from a point on a circle or terminal side.

I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED

pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

UNIT LEARNING OBJECTIVES

I can graph basic sine and cosine functions using key features.

I can identify amplitude, midline, period, and phase shift from an equation or graph.

I can graph transformed sine and cosine functions.

I can write sine and cosine equations that represent a periodic graph.

I can construct a sinusoidal model from contextual information or data.

I can interpret the parameters and outputs of a sinusoidal model in context.

I can graph tangent, secant, and cosecant functions with their asymptotes.

I can determine whether a function is even, odd, or neither from its graph or formula.

Inverse Functions and Trigonometric Equations

REQUIRED

pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

UNIT LEARNING OBJECTIVES

I can read inverse-trigonometric notation and distinguish it from reciprocal notation.

I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

I can explain how restricted domains make inverse trigonometric functions possible.

I can solve trigonometric equations exactly or approximately on a specified interval.

I can express all solutions of a trigonometric equation in general form.

I can evaluate compositions of trigonometric and inverse trigonometric functions.

Non-Right Triangles

REQUIRED

pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

UNIT LEARNING OBJECTIVES

I can solve a non-right triangle using the Law of Sines.

I can determine the number of possible triangles in the ambiguous SSA case.

I can solve a non-right triangle using the Law of Cosines.

I can select and apply an appropriate method to solve a triangle in context.

Trigonometric Identities

REQUIRED

pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

UNIT LEARNING OBJECTIVES

I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.

I can rewrite trigonometric expressions using the Pythagorean identities.

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSION

pp. 117-118, 124-126 Double-angle and sum/difference identities

UNIT LEARNING OBJECTIVES

I can apply double-angle identities.

I can apply half-angle identities.

I can apply angle sum and difference identities.

I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSION

pp. 101 New Area Formula

UNIT LEARNING OBJECTIVES

I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED

pp. 127-128 Polar Coordinates and Coordinate Conversion

UNIT LEARNING OBJECTIVES

I can plot and identify points in polar coordinates.

I can represent a polar point using equivalent coordinate pairs.

I can convert points between polar and rectangular coordinates.

Polar Functions

EXTENSION

pp. 129-132 Polar Functions; pp. 133-135 Polar/Rectangular Function Conversion

UNIT LEARNING OBJECTIVES

I can graph and interpret basic polar functions.

I can convert equations between polar and rectangular forms.

Vectors

EXTENSION

pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

UNIT LEARNING OBJECTIVES

I can represent a vector geometrically and in component form.

I can add, subtract, and multiply vectors by scalars.

I can determine vector magnitude, direction, and unit vector.

I can resolve a vector into horizontal and vertical components.

I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSION

pp. 145-147 Introduction to Parametrics

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Trigonometric Foundations

REQUIRED

Resource coverage: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

M32-FND-001

I can use the Pythagorean theorem to determine an unknown side of a right triangle.

Resource: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- R Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22

M32-FND-002

I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

Resource: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Distinguish exact values from measured or rounded approximations.
  SK-GEO-EXACT-APPROX · first introduced in Math 22
- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- R Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS · first introduced in Math 22

M32-FND-003

I can convert angle measures between degrees and radians.

Resource: pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- I Convert between degree and radian measure.
  SK-TRIG-ANGLE-CONVERT

M32-FND-004

I can determine coterminal and reference angles for a given angle.

Resource: pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 29-33 Review Questions

SUPPORTING SKILLS

- I Generate and recognize coterminal angles.
  SK-TRIG-ANGLE-COTERMINAL
- I Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE
- I Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT

M32-FND-005

I can calculate arc length from a central angle and radius.

Resource: pp. 13-14 Arc Length; pp. 29-33 Review Questions

SUPPORTING SKILLS

- D Calculate arc length from radius and central angle.
  SK-CIR-ARC-LENGTH · first introduced in Math 22
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22
- A Convert between degree and radian measure.
  SK-TRIG-ANGLE-CONVERT · first introduced in Math 32

M32-FND-006

I can determine exact coordinates on the unit circle.

Resource: pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS · first introduced in Math 22
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- I Determine unit-circle coordinates for common angles.
  SK-TRIG-UNIT-CIRCLE-COORD

M32-FND-007

I can evaluate all six trigonometric functions exactly for common angles.

Resource: pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 29-33 Review Questions; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions

SUPPORTING SKILLS

- I Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE
- I Recall the six trigonometric functions, their reciprocal relationships, and their notation.
  SK-TRIG-NOTATION-VOCAB
- I Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL
- A Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT · first introduced in Math 32
- A Determine unit-circle coordinates for common angles.
  SK-TRIG-UNIT-CIRCLE-COORD · first introduced in Math 32

M32-FND-008

I can solve right triangles using trigonometric ratios.

Resource: pp. 22-23 Right Triangle Applications; pp. 29-33 Review Questions

SUPPORTING SKILLS

- D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- D Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- D Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22
- R Label sides relative to a selected angle.
  SK-TRIG-TRIANGLE-LABEL · first introduced in Math 22

M32-FND-009

I can apply right-triangle trigonometry to contextual problems.

Resource: pp. 22-23 Right Triangle Applications; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12
- D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- A Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- A Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22

M32-FND-010

I can evaluate trigonometric functions from a point on a circle or terminal side.

Resource: pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22
- I Use coordinates and radius to evaluate trigonometric functions.
  SK-TRIG-CIRCLE-POINT
- A Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL · first introduced in Math 32
- A Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT · first introduced in Math 32

M32-FND-011

I can solve a basic trigonometric equation for all angles in a specified interval.

Resource: pp. 34-45 Multiple Angle Solutions; Reciprocal Functions

SUPPORTING SKILLS

- A Translate between inequality and interval notation.
  SK-NOT-INTERVAL · first introduced in Math 12
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- I Use reference angles and periodicity to find every solution in an interval.
  SK-TRIG-EQUATION-INTERVAL
- I Isolate a trigonometric expression before solving.
  SK-TRIG-EQUATION-ISOLATE
- A Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT · first introduced in Math 32

Periodic Functions and Modeling

REQUIRED

Resource coverage: pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

M32-GRF-001

I can graph basic sine and cosine functions using key features.

Resource: pp. 46-49 Graphing Sine and Cosine

SUPPORTING SKILLS

- D Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH · first introduced in Math 21
- I Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS
- A Determine unit-circle coordinates for common angles.
  SK-TRIG-UNIT-CIRCLE-COORD · first introduced in Math 32

M32-GRF-002

I can identify amplitude, midline, period, and phase shift from an equation or graph.

Resource: pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- I Determine amplitude and vertical reflection.
  SK-TRIG-GRAPH-AMPLITUDE
- I Determine a sinusoid's midline and vertical shift.
  SK-TRIG-GRAPH-MIDLINE
- I Determine period from a graph or equation.
  SK-TRIG-GRAPH-PERIOD
- I Determine horizontal shift from a graph or equation.
  SK-TRIG-GRAPH-PHASE
- I Recall and correctly use the terms amplitude, midline, period, phase shift, vertical shift, and asymptote.
  SK-TRIG-GRAPH-VOCAB

M32-GRF-003

I can graph transformed sine and cosine functions.

Resource: pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- A Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- D Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32

M32-GRF-004

I can write sine and cosine equations that represent a periodic graph.

Resource: pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D Determine amplitude and vertical reflection.
  SK-TRIG-GRAPH-AMPLITUDE · first introduced in Math 32
- D Determine a sinusoid's midline and vertical shift.
  SK-TRIG-GRAPH-MIDLINE · first introduced in Math 32
- D Determine period from a graph or equation.
  SK-TRIG-GRAPH-PERIOD · first introduced in Math 32
- D Determine horizontal shift from a graph or equation.
  SK-TRIG-GRAPH-PHASE · first introduced in Math 32

M32-GRF-005

I can construct a sinusoidal model from contextual information or data.

Resource: pp. 69-76 Modeling Problems

SUPPORTING SKILLS

- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Determine amplitude and vertical reflection.
  SK-TRIG-GRAPH-AMPLITUDE · first introduced in Math 32
- A Determine a sinusoid's midline and vertical shift.
  SK-TRIG-GRAPH-MIDLINE · first introduced in Math 32
- A Determine period from a graph or equation.
  SK-TRIG-GRAPH-PERIOD · first introduced in Math 32
- I Translate contextual maximum, minimum, cycle, and starting position into model parameters.
  SK-TRIG-MODEL-PARAMETERS

M32-GRF-006

I can interpret the parameters and outputs of a sinusoidal model in context.

Resource: pp. 69-76 Modeling Problems

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12
- D Translate contextual maximum, minimum, cycle, and starting position into model parameters.
  SK-TRIG-MODEL-PARAMETERS · first introduced in Math 32

M32-GRF-007

I can graph tangent, secant, and cosecant functions with their asymptotes.

Resource: pp. 77-85 Graphing Cosecant, Secant, and Tangent

SUPPORTING SKILLS

- **A** Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- **I** Locate vertical asymptotes of tangent, secant, and cosecant.
  SK-TRIG-GRAPH-ASYMPTOTE
- **A** Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32
- **D** Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL · first introduced in Math 32

M32-GRF-008

I can determine whether a function is even, odd, or neither from its graph or formula.

Resource: pp. 46-49 Graphing Sine and Cosine

SUPPORTING SKILLS

- **D** Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY · first introduced in Math 31

Inverse Functions and Trigonometric Equations

REQUIRED

Resource coverage: pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

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M32-INV-006

I can read inverse-trigonometric notation and distinguish it from reciprocal notation.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Read inverse-trigonometric notation and distinguish it from reciprocal notation.
  SK-TRIG-INVERSE-NOTATION
- A — Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL · first introduced in Math 32

M32-INV-001

I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

Resource: pp. 86-91 Calculating Angles; Inverse Trig Functions

SUPPORTING SKILLS

- A — Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE · first introduced in Math 32
- I — Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE
- A — Read inverse-trigonometric notation and distinguish it from reciprocal notation.
  SK-TRIG-INVERSE-NOTATION · first introduced in Math 32
- I — Recall the principal-value ranges of inverse trigonometric functions.
  SK-TRIG-INVERSE-RANGE

M32-INV-002

I can explain how restricted domains make inverse trigonometric functions possible.

Resource: pp. 86-91 Calculating Angles; Inverse Trig Functions

SUPPORTING SKILLS

- A — Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- A — Apply the horizontal line test for invertibility.
  SK-FUNC-INVERSE-HLT · first introduced in Math 31
- A — Test whether a function is one-to-one on a domain.
  SK-FUNC-ONE-TO-ONE · first introduced in Math 31
- D — Recall the principal-value ranges of inverse trigonometric functions.
  SK-TRIG-INVERSE-RANGE · first introduced in Math 32

M32-INV-003

I can solve trigonometric equations exactly or approximately on a specified interval.

Resource: pp. 92-98 Solving Trig Equations, Part 2

SUPPORTING SKILLS

- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- D Use reference angles and periodicity to find every solution in an interval.
  SK-TRIG-EQUATION-INTERVAL · first introduced in Math 32
- D Isolate a trigonometric expression before solving.
  SK-TRIG-EQUATION-ISOLATE · first introduced in Math 32
- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32

M32-INV-004

I can express all solutions of a trigonometric equation in general form.

Resource: pp. 92-98 Solving Trig Equations, Part 2

SUPPORTING SKILLS

- D Generate and recognize coterminal angles.
  SK-TRIG-ANGLE-COTERMINAL · first introduced in Math 32
- I Express a periodic solution family in general form.
  SK-TRIG-EQUATION-GENERAL
- A Use reference angles and periodicity to find every solution in an interval.
  SK-TRIG-EQUATION-INTERVAL · first introduced in Math 32

M32-INV-005

I can evaluate compositions of trigonometric and inverse trigonometric functions.

Resource: pp. 99-100 Compositions of Trig and Inverse Trig Functions

SUPPORTING SKILLS

- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- A Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- A Recall the principal-value ranges of inverse trigonometric functions.
  SK-TRIG-INVERSE-RANGE · first introduced in Math 32

Non-Right Triangles

REQUIRED

Resource coverage: pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

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M32-TRI-002

I can solve a non-right triangle using the Law of Sines.

Resource: pp. 102-110 Law of Sines; Ambiguous Case

SUPPORTING SKILLS

- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- I Apply the Law of Sines to determine missing sides or angles.
  SK-TRIG-LAW-SINES
- D Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-003

I can determine the number of possible triangles in the ambiguous SSA case.

Resource: pp. 102-110 Law of Sines; Ambiguous Case

SUPPORTING SKILLS

- I Test and interpret the possible outcomes of SSA data.
  SK-TRIG-AMBIGUOUS-CASE
- D Apply the Law of Sines to determine missing sides or angles.
  SK-TRIG-LAW-SINES · first introduced in Math 32
- A Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-004

I can solve a non-right triangle using the Law of Cosines.

Resource: pp. 111-115 Law of Cosines; Mixed Practice

SUPPORTING SKILLS

- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- I Apply the Law of Cosines to determine missing sides or angles.
  SK-TRIG-LAW-COSINES
- D Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-005

I can select and apply an appropriate method to solve a triangle in context.

Resource: pp. 111-115 Law of Cosines; Mixed Practice

SUPPORTING SKILLS

- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12
- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- A Apply the Law of Cosines to determine missing sides or angles.
  SK-TRIG-LAW-COSINES · first introduced in Math 32
- A Apply the Law of Sines to determine missing sides or angles.
  SK-TRIG-LAW-SINES · first introduced in Math 32
- D Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32
- I Select right-triangle ratios, Law of Sines, or Law of Cosines from the given data.
  SK-TRIG-TRIANGLE-METHOD

Trigonometric Identities

REQUIRED

Resource coverage: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

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M32-IDN-001

I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- I — Apply reciprocal and quotient identities.
  SK-TRIG-IDENTITY-RECIPROCAL-QUOTIENT
- I — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE

M32-IDN-002

I can rewrite trigonometric expressions using the Pythagorean identities.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- I — Apply the Pythagorean trigonometric identities.
  SK-TRIG-IDENTITY-PYTHAGOREAN
- D — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDN-003

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- D — Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY · first introduced in Math 31
- I — Apply sine/cosine cofunction identities.
  SK-TRIG-IDENTITY-COFUNCTION
- I — Apply even and odd trigonometric identities.
  SK-TRIG-IDENTITY-EVEN-ODD
- D — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDN-004

I can verify a trigonometric identity using the required identity families.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I — Choose a productive identity or algebraic manipulation.
  SK-TRIG-IDENTITY-STRATEGY
- D — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32
- I — Maintain equivalence while transforming one side of an identity.
  SK-TRIG-IDENTITY-VERIFY

Additional Trigonometric Identities

EXTENSION

Resource coverage: pp. 117-118, 124-126 Double-angle and sum/difference identities

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M32-IDX-001

I can apply double-angle identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- I — Apply double-angle identities.
  SK-TRIG-IDENTITY-DOUBLE-ANGLE
- A — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDX-002

I can apply half-angle identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- A — Apply double-angle identities.
  SK-TRIG-IDENTITY-DOUBLE-ANGLE · first introduced in Math 32
- I — Apply half-angle identities.
  SK-TRIG-IDENTITY-HALF-ANGLE

M32-IDX-003

I can apply angle sum and difference identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- A — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32
- I — Apply angle sum and difference identities.
  SK-TRIG-IDENTITY-SUM-DIFFERENCE

M32-IDX-004

I can determine exact trigonometric values using extension identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- D — Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE · first introduced in Math 32
- A — Apply double-angle identities.
  SK-TRIG-IDENTITY-DOUBLE-ANGLE · first introduced in Math 32
- A — Apply half-angle identities.
  SK-TRIG-IDENTITY-HALF-ANGLE · first introduced in Math 32
- A — Apply angle sum and difference identities.
  SK-TRIG-IDENTITY-SUM-DIFFERENCE · first introduced in Math 32

Triangle Area from SAS

EXTENSION

Resource coverage: pp. 101 New Area Formula

M32-TRX-001

I can calculate the area of a triangle from two sides and the included angle.

Resource: pp. 101 New Area Formula

SUPPORTING SKILLS

- A Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE · first introduced in Math 32
- I Calculate triangle area from two sides and the included angle.
  SK-TRIG-TRIANGLE-AREA-SAS
- I Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA

Polar Coordinates REQUIRED

Resource coverage: pp. 127-128 Polar Coordinates and Coordinate Conversion

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M32-POL-001

I can plot and identify points in polar coordinates.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I Read and write polar-coordinate notation and identify the radius, angle, pole, and polar axis.
  SK-POL-NOTATION-READ
- I Plot a point from radius and angle.
  SK-POL-PLOT
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32

M32-POL-002

I can represent a polar point using equivalent coordinate pairs.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I Generate equivalent polar coordinate pairs.
  SK-POL-EQUIVALENT
- A Generate and recognize coterminal angles.
  SK-TRIG-ANGLE-COTERMINAL · first introduced in Math 32

M32-POL-003

I can convert points between polar and rectangular coordinates.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I Use coordinate relationships to convert a point between systems.
  SK-POL-CONVERT-POINT
- A Use coordinates and radius to evaluate trigonometric functions.
  SK-TRIG-CIRCLE-POINT · first introduced in Math 32
- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32

Polar Functions

EXTENSION

Resource coverage: pp. 129-132 Polar Functions; pp. 133-135 Polar/Rectangular Function Conversion

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M32-PLX-001

I can graph and interpret basic polar functions.

Resource: pp. 129-132 Polar Functions

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Plot a point from radius and angle.
  SK-POL-PLOT · first introduced in Math 32
- A Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32

M32-PLX-002

I can convert equations between polar and rectangular forms.

Resource: pp. 133-135 Polar/Rectangular Function Conversion

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Substitute coordinate identities to convert an equation between systems.
  SK-POL-CONVERT-EQUATION
- A Apply the Pythagorean trigonometric identities.
  SK-TRIG-IDENTITY-PYTHAGOREAN · first introduced in Math 32

Vectors EXTENSION

Resource coverage: pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

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M32-VEC-001

I can represent a vector geometrically and in component form.

Resource: pp. 136-140 Vector Concepts and Operations

SUPPORTING SKILLS

- A Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK · first introduced in Math 22
- I Write a vector in component form.
  SK-VEC-COMPONENTS
- I Read and write vector notation and correctly use the terms magnitude, direction, component, and unit vector.
  SK-VEC-NOTATION-VOCAB

M32-VEC-002

I can add, subtract, and multiply vectors by scalars.

Resource: pp. 136-140 Vector Concepts and Operations

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide signed integers.
  SK-NUM-INTEGER-OPERATE · inherited from middle school
- I Perform component-wise vector operations.
  SK-VEC-OPERATE

M32-VEC-003

I can determine vector magnitude, direction, and unit vector.

Resource: pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22
- I Determine a vector's direction angle with correct quadrant.
  SK-VEC-DIRECTION
- I Calculate vector magnitude using the distance formula.
  SK-VEC-MAGNITUDE
- I Normalize a nonzero vector to obtain a unit vector.
  SK-VEC-UNIT

M32-VEC-004

I can resolve a vector into horizontal and vertical components.

Resource: pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- D Write a vector in component form.
  SK-VEC-COMPONENTS · first introduced in Math 32
- A Determine a vector's direction angle with correct quadrant.
  SK-VEC-DIRECTION · first introduced in Math 32

M32-VEC-005

I can perform basic operations with vectors in three dimensions.

Resource: pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- I Represent and operate on vectors with three components.
  SK-VEC-3D
- D Calculate vector magnitude using the distance formula.
  SK-VEC-MAGNITUDE · first introduced in Math 32
- D Perform component-wise vector operations.
  SK-VEC-OPERATE · first introduced in Math 32

Parametric Equations

EXTENSION

Resource coverage: pp. 145-147 Introduction to Parametrics

M32-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH
- I Read parametric notation and correctly use the terms parameter, coordinate equation, orientation, and parameter interval.
  SK-PAR-NOTATION-VOCAB
- I Generate ordered pairs from parametric equations.
  SK-PAR-TABLE

M32-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I Eliminate a parameter algebraically.
  SK-PAR-ELIMINATE
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M32-PAR-003

I can create or interpret parametric equations for motion in the plane.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I Interpret or construct parametric equations for planar motion.
  SK-PAR-MODEL
- D Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Math 39 Curriculum

Precalculus with Data Analysis | Middlesex School Mathematics | Resource: Math 39

COURSE AT A GLANCE

Linear Modeling and Residuals

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can calculate and interpret average rate of change with appropriate units.

I can construct a linear function from two data points, a table, a graph, or a context.

I can fit a linear regression model to data using technology and interpret its parameters.

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

I can construct an exponential function from two data points, a table, or a context.

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

I can determine and interpret doubling time or half-life from an exponential model.

I can solve exponential and logarithmic equations in modeling contexts.

I can fit an exponential regression model using technology and evaluate its fit.

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can identify a power function and interpret the parameters in y = kx^p.

I can construct a power function from two data points or a proportional relationship.

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

I can recognize and model direct, inverse, and other power-variation relationships.

I can solve equations involving integer or rational powers.

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

I can determine where a combination of functions is positive, zero, or undefined.

I can determine polynomial end behavior from degree and leading coefficient.

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

I can construct a polynomial function from specified zeros, multiplicities, and a point.

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

I can build and optimize a polynomial model in a geometric or economic context.

Describing Data

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can distinguish categorical and quantitative data.

I can construct and interpret frequency and relative-frequency tables.

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

I can identify misleading features in a data display.

I can describe a distribution's modality, symmetry, skewness, and outliers.

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

I can construct and interpret a five-number summary and boxplot.

I can use percentiles and z-scores to locate and compare values within distributions.

I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can interpret and write notation for compound and conditional probabilities.

I can organize categorical data in a contingency table or Venn diagram.

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

I can identify an experiment, outcomes, events, and a sample space.

I can calculate theoretical probabilities when outcomes are equally likely.

I can use complements to calculate probabilities.

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can calculate the expected value of a discrete probability situation.

I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can linearize power or exponential data and translate between the linearized and original models.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Linear Modeling and Residuals

REQUIRED

Resource coverage: No mapped textbook section

M39-LIN-001

I can calculate and interpret average rate of change with appropriate units.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Calculate average rate of change over an interval.
  SK-FUNC-AROC · first introduced in Math 21
- D Interpret average rate of change with contextual units.
  SK-FUNC-AROC-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M39-LIN-002

I can construct a linear function from two data points, a table, a graph, or a context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- R Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12
- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-LIN-003

I can fit a linear regression model to data using technology and interpret its parameters.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I Use technology to determine a least-squares linear regression model.
  SK-REG-LINEAR-FIT
- I Recall and correctly use regression terms including explanatory variable, response variable, correlation, residual, and residual plot.
  SK-REG-VOCABULARY

M39-LIN-004

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Interpret the sign and magnitude of a correlation coefficient in context.
  SK-REG-CORRELATION-INTERPRET

M39-LIN-005

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Use residual patterns and contextual evidence to judge whether a model is appropriate.
  SK-REG-MODEL-ASSESS
- I Calculate an observed value minus its model-predicted value.
  SK-REG-RESIDUAL-CALCULATE
- I Construct and interpret a residual plot.
  SK-REG-RESIDUAL-PLOT

M39-LIN-006

I can use a linear model to make and interpret predictions, including solving for either variable.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12

Exponential Modeling

REQUIRED

Resource coverage: No mapped textbook section

M39-EXP-001

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Use constant differences or ratios to distinguish linear and exponential patterns.
  SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31
- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21

M39-EXP-002

I can construct an exponential function from two data points, a table, or a context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- D Determine an exponential model from tabular values.
  SK-EXP-MODEL-TABLE · first introduced in Math 31
- D Solve for an exponential model using two points.
  SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31

M39-EXP-003

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Convert a percent increase or decrease to a growth or decay factor.
  SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- R Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
  SK-EXP-VOCABULARY · first introduced in Math 31
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-EXP-004

I can determine and interpret doubling time or half-life from an exponential model.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- D Translate a doubling-time or half-life question into an exponential equation.
  SK-LOG-DOUBLING-HALF · first introduced in Math 31

M39-EXP-005

I can solve exponential and logarithmic equations in modeling contexts.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- R Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- R Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31

M39-EXP-006

I can fit an exponential regression model using technology and evaluate its fit.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Interpret the sign and magnitude of a correlation coefficient in context.
  SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I Use technology to determine an exponential regression model.
  SK-REG-EXPONENTIAL-FIT
- D Use residual patterns and contextual evidence to judge whether a model is appropriate.
  SK-REG-MODEL-ASSESS · first introduced in Math 39

M39-EXP-007

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- I Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT
- D Use residual patterns and contextual evidence to judge whether a model is appropriate.
  SK-REG-MODEL-ASSESS · first introduced in Math 39

Power Functions and Function Comparison

REQUIRED

Resource coverage: No mapped textbook section

M39-POW-001

I can identify a power function and interpret the parameters in y = kx^p.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Interpret the coefficient and exponent in a power function.
  SK-POWER-PARAMETER-INTERPRET
- I — Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
  SK-POWER-VOCABULARY
- I — Represent and interpret a power-variation relationship.
  SK-VAR-POWER

M39-POW-002

I can construct a power function from two data points or a proportional relationship.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A — Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I — Determine a power function from two points or a proportional relationship.
  SK-POWER-CONSTRUCT

M39-POW-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Determine power-function behavior as the input approaches zero or positive or negative infinity.
  SK-POWER-END-BEHAVIOR
- I — Relate the coefficient and exponent of a power function to its graph shape.
  SK-POWER-SHAPE
- D — Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M39-POW-004

I can recognize and model direct, inverse, and other power-variation relationships.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Represent and interpret a direct-variation relationship.
  SK-VAR-DIRECT
- I — Represent and interpret an inverse-variation relationship.
  SK-VAR-INVERSE
- D — Represent and interpret a power-variation relationship.
  SK-VAR-POWER · first introduced in Math 39

M39-POW-005

I can solve equations involving integer or rational powers.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21

M39-POW-006

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- I Compare the long-run growth or decay of functions from different families.
  SK-FUNC-DOMINANCE

M39-POW-007

I can fit a power regression model and interpret its parameters and correlation evidence.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Interpret the coefficient and exponent in a power function.
  SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- A Interpret the sign and magnitude of a correlation coefficient in context.
  SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I Use technology to determine a power regression model.
  SK-REG-POWER-FIT

Function Combinations and Polynomials REQUIRED

Resource coverage: No mapped textbook section

M39-POL-001

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Add, subtract, multiply, and divide function outputs.
  SK-FUNC-OPERATIONS

M39-POL-002

I can determine where a combination of functions is positive, zero, or undefined.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Determine the domain of a sum, difference, product, quotient, or composition.
  SK-FUNC-DOMAIN-COMBINE
- D Add, subtract, multiply, and divide function outputs.
  SK-FUNC-OPERATIONS · first introduced in Math 39

M39-POL-003

I can determine polynomial end behavior from degree and leading coefficient.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR
- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M39-POL-004

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12
- D Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M39-POL-005

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Use inflection points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-INFLECTION
- I Use turning points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-TURNING
- I Use zeros and their multiplicities to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-ZEROS
- D Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39

M39-POL-006

I can construct a polynomial function from specified zeros, multiplicities, and a point.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT · first introduced in Math 31
- A Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M39-POL-007

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39
- A Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M39-POL-008

I can build and optimize a polynomial model in a geometric or economic context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES · first introduced in Math 21

Describing Data

REQUIRED

Resource coverage: No mapped textbook section

M39-STA-001

I can distinguish categorical and quantitative data.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Classify data as categorical or quantitative.
  SK-STAT-DATA-TYPE

M39-STA-002

I can construct and interpret frequency and relative-frequency tables.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Construct and interpret a frequency table.
  SK-STAT-FREQUENCY-TABLE
- I Calculate and interpret relative frequency.
  SK-STAT-RELATIVE-FREQUENCY

M39-STA-003

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Construct and interpret bar charts and pie charts.
  SK-STAT-CATEGORICAL-DISPLAY
- I Construct and interpret a histogram for quantitative data.
  SK-STAT-HISTOGRAM

M39-STA-004

I can identify misleading features in a data display.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Identify perceptual distortion, misleading scales, and other misleading features in data displays.
  SK-STAT-GRAPH-CRITIQUE

M39-STA-005

I can describe a distribution's modality, symmetry, skewness, and outliers.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Describe a distribution by modality, symmetry, and skewness.
  SK-STAT-DISTRIBUTION-SHAPE
- I Identify and interpret unusual observations or outliers.
  SK-STAT-OUTLIER-IDENTIFY
- I Recall and correctly use terms for data type, distribution shape, center, spread, percentile, and outlier.
  SK-STAT-VOCABULARY

M39-STA-006

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Select an appropriate measure of center based on distribution shape and context.
  SK-STAT-CENTER-SELECT
- I Calculate and interpret the arithmetic mean.
  SK-STAT-MEAN
- I Calculate and interpret the median.
  SK-STAT-MEDIAN
- I Identify and interpret the mode or modes.
  SK-STAT-MODE

M39-STA-007

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Determine quartiles and calculate and interpret the interquartile range.
  SK-STAT-QUARTILES-IQR
- I Calculate and interpret the range.
  SK-STAT-RANGE
- I Calculate and interpret standard deviation as a measure of spread.
  SK-STAT-STANDARD-DEVIATION

M39-STA-008

I can construct and interpret a five-number summary and boxplot.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Construct and interpret a boxplot.
  SK-STAT-BOXPLOT
- I Determine and interpret a five-number summary.
  SK-STAT-FIVE-NUMBER

M39-STA-009

I can use percentiles and z-scores to locate and compare values within distributions.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Determine and interpret the percentile rank of a value.
  SK-STAT-PERCENTILE
- I Calculate and interpret a z-score.
  SK-STAT-ZSCORE

M39-STA-010

I can compare distributions using their shape, center, spread, and unusual features.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Select an appropriate measure of center based on distribution shape and context.
  SK-STAT-CENTER-SELECT · first introduced in Math 39
- I Compare distributions using shape, center, spread, and unusual features.
  SK-STAT-DISTRIBUTION-COMPARE
- A Describe a distribution by modality, symmetry, and skewness.
  SK-STAT-DISTRIBUTION-SHAPE · first introduced in Math 39

Probability

REQUIRED

Resource coverage: No mapped textbook section

M39-PRO-009

I can interpret and write notation for compound and conditional probabilities.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.
  SK-PROB-NOTATION-VOCAB

M39-PRO-001

I can organize categorical data in a contingency table or Venn diagram.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Construct and interpret a contingency table.
  SK-PROB-CONTINGENCY-TABLE
- I — Represent categorical events and their overlap with a Venn diagram.
  SK-PROB-VENN-DIAGRAM

M39-PRO-002

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Calculate and interpret a conditional probability.
  SK-PROB-CONDITIONAL
- I — Calculate probability from observed relative frequency.
  SK-PROB-EMPIRICAL
- I — Calculate and interpret marginal and joint probabilities.
  SK-PROB-MARGINAL-JOINT

M39-PRO-003

I can identify an experiment, outcomes, events, and a sample space.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A — Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.
  SK-PROB-NOTATION-VOCAB · first introduced in Math 39
- I — Identify outcomes, events, and the sample space of an experiment.
  SK-PROB-SAMPLE-SPACE

M39-PRO-004

I can calculate theoretical probabilities when outcomes are equally likely.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A Identify outcomes, events, and the sample space of an experiment.
  SK-PROB-SAMPLE-SPACE · first introduced in Math 39
- I Calculate theoretical probability for equally likely outcomes.
  SK-PROB-THEORETICAL

M39-PRO-005

I can use complements to calculate probabilities.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Use the complement rule to calculate probability.
  SK-PROB-COMPLEMENT

M39-PRO-006

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Calculate the probability of a compound and event.
  SK-PROB-AND
- D Calculate and interpret a conditional probability.
  SK-PROB-CONDITIONAL · first introduced in Math 39
- I Determine whether events are independent or dependent.
  SK-PROB-INDEPENDENCE

M39-PRO-007

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Determine whether events are overlapping or disjoint.
  SK-PROB-DISJOINT
- I Calculate the probability of a compound or event.
  SK-PROB-OR

M39-PRO-008

I can compare experimental and theoretical probability and explain the law of large numbers.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D Calculate probability from observed relative frequency.
  SK-PROB-EMPIRICAL · first introduced in Math 39
- I Explain how experimental probability stabilizes with many trials.
  SK-PROB-LARGE-NUMBERS
- D Calculate theoretical probability for equally likely outcomes.
  SK-PROB-THEORETICAL · first introduced in Math 39

Expected Value

EXTENSION

Resource coverage: No mapped textbook section

M39-EV-001

I can calculate the expected value of a discrete probability situation.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Read expected-value notation and recall the meanings of outcome, probability, payoff, and long-run average.
  SK-PROB-EXPECTED-NOTATION
- I — Calculate the expected value of a discrete probability situation.
  SK-PROB-EXPECTED-VALUE

M39-EV-002

I can interpret expected value to evaluate a game, risk, or long-run decision.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Interpret expected value as a long-run average for decision-making.
  SK-PROB-EXPECTED-INTERPRET
- D — Calculate the expected value of a discrete probability situation.
  SK-PROB-EXPECTED-VALUE · first introduced in Math 39

Linearization of Nonlinear Data EXTENSION

Resource coverage: No mapped textbook section

M39-POW-008

I can linearize power or exponential data and translate between the linearized and original models.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I Transform exponential data so that a linear model can be fit to its logarithms.
  SK-REG-LOG-LINEARIZE
- I Transform power data so that a linear model can be fit on logarithmic axes.
  SK-REG-LOGLOG-LINEARIZE
- I Translate a linearized regression equation back to its original nonlinear form.
  SK-REG-UNLINEARIZE

Math 49 Curriculum

Precalculus with Limits | Middlesex School Mathematics | Textbook: OpenStax Precalculus 2e (Jay Abramson, 2021); OpenStax Calculus Volume 1, Chapter 2, "Limits" (Gilbert Strang and Edwin Herman)

COURSE AT A GLANCE

Review Content

REVIEW

1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties

UNIT LEARNING OBJECTIVES

I can determine the domain and range of a function from a formula, graph, or table.

I can describe and apply transformations of a parent function.

I can analyze exponential functions using their equations, graphs, and contexts.

I can analyze logarithmic functions using their equations, graphs, and contexts.

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED

1.4 Composition of Functions; 1.7 Inverse Functions

UNIT LEARNING OBJECTIVES

I can determine whether a function is even, odd, or neither.

I can add, subtract, multiply, and divide functions and state the resulting domain.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for the composition of two functions and determine its domain.

I can decompose a function into a composition of simpler functions.

I can determine whether a function has an inverse on a given domain.

I can find and verify an inverse function algebraically.

I can interpret the relationship between a function and its inverse from graphs or tables.

Power Functions

REQUIRED

3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation; 3.8 Inverses and Radical Functions

UNIT LEARNING OBJECTIVES

I can identify a power function and interpret the parameters in y = kx^p.

I can construct a power function from two data points or a proportional relationship.

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

I can recognize and model direct, inverse, and other power-variation relationships.

I can solve equations involving integer or rational powers.

I can build or select a power-function model for a context and interpret its parameters.

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Polynomial Functions

REQUIRED

3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

UNIT LEARNING OBJECTIVES

I can predict the end behavior of a polynomial from its degree and leading coefficient.

I can determine polynomial zeros and their multiplicities.

I can construct a polynomial function from specified zeros or graphical features.

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

I can build or select a polynomial model for a context and interpret its features.

Rational Functions and Limits

REQUIRED

3.7 Rational Functions; 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.3 The Limit Laws; 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 4.4 Graphs of Logarithmic Functions; Chapter 2 Limits; 3.9 Modeling Using Variation

UNIT LEARNING OBJECTIVES

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

I can sketch and analyze a rational function using its algebraic and asymptotic features.

I can interpret limit notation as a statement about function behavior near an input.

I can estimate a limit from a graph or table.

I can evaluate a limit algebraically using direct substitution or simplification.

I can interpret an infinite limit as vertical-asymptotic behavior.

I can use limit notation to describe behavior at infinity.

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Sequences and Series

REQUIRED

11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

UNIT LEARNING OBJECTIVES

I can evaluate and graph a sequence defined explicitly or recursively.

I can write explicit and recursive formulas for arithmetic sequences.

I can write explicit and recursive formulas for geometric sequences.

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

I can create and interpret a sequence model for discrete change.

I can interpret and use summation notation.

I can calculate a finite arithmetic series.

I can calculate a finite geometric series.

I can determine whether an infinite geometric series converges and find its sum when it does.

I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.4 Continuity

UNIT LEARNING OBJECTIVES

I can determine one-sided limits and decide whether a two-sided limit exists.

I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

8.7 Parametric Equations: Graphs; 8.6 Parametric Equations

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates; 10.3 The Parabola

UNIT LEARNING OBJECTIVES

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

I can use completing the square to rewrite a general-form conic equation in standard form.

I can write, interpret, and graph implicit and parametric equations of a circle.

I can write, interpret, and graph implicit and parametric equations of an ellipse.

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

Conic Sections (Continued)

EXTENSION

UNIT LEARNING OBJECTIVES - CONTINUED

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

I can explain a parabola as the set of points equidistant from a focus and directrix.

IN-DEPTH CURRICULUM

SKILL PROGRESSION — I Introduce — R Reinforce — D Deepen — A Apply

Review Content

REVIEW

Textbook coverage: 1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties

M49-FND-001

I can determine the domain and range of a function from a formula, graph, or table.

Textbook: 1.2 Domain and Range

SUPPORTING SKILLS

- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- R Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- R Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M49-FND-002

I can describe and apply transformations of a parent function.

Textbook: 1.5 Transformation of Functions

SUPPORTING SKILLS

- R Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- R Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- R Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M49-FND-004

I can analyze exponential functions using their equations, graphs, and contexts.

Textbook: 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- R Identify horizontal asymptotic behavior of an exponential graph.
  SK-EXP-ASYMPTOTE · first introduced in Math 31
- R Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-FND-005

I can analyze logarithmic functions using their equations, graphs, and contexts.

Textbook: 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- R Identify vertical asymptotic behavior of a logarithmic graph.
  SK-LOG-ASYMPTOTE · first introduced in Math 31
- R Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-FND-006

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Textbook: 4.3 Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- R Rewrite exponential expressions using a common base when possible.
  SK-EXP-REWRITE-BASE · first introduced in Math 31
- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A Apply the logarithm power property in either direction.
  SK-LOG-POWER · first introduced in Math 31
- A Apply the logarithm product property in either direction.
  SK-LOG-PRODUCT · first introduced in Math 31
- A Apply the logarithm quotient property in either direction.
  SK-LOG-QUOTIENT · first introduced in Math 31
- R Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- R Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31

Function Operations, Composition, and Inverses REQUIRED

Textbook coverage: 1.4 Composition of Functions; 1.7 Inverse Functions

M49-FND-003

I can determine whether a function is even, odd, or neither.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- D Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY · first introduced in Math 31

M49-OPS-001

I can add, subtract, multiply, and divide functions and state the resulting domain.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Determine the domain of a sum, difference, product, quotient, or composition.
  SK-FUNC-DOMAIN-COMBINE
- I Add, subtract, multiply, and divide function outputs.
  SK-FUNC-OPERATIONS

M49-OPS-002

I can evaluate a composition from formulas, graphs, or tables.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Follow intermediate values to compose functions from tables.
  SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- R Read and write notation for function composition and inverse functions.
  SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M49-OPS-003

I can write a formula for the composition of two functions and determine its domain.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Determine the domain of a sum, difference, product, quotient, or composition.
  SK-FUNC-DOMAIN-COMBINE · first introduced in Math 39

M49-OPS-004

I can decompose a function into a composition of simpler functions.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I Identify simpler inner and outer functions in a composition.
  SK-FUNC-DECOMPOSE

M49-OPS-005

I can determine whether a function has an inverse on a given domain.

Textbook: 1.7 Inverse Functions

SUPPORTING SKILLS

- D Apply the horizontal line test for invertibility.
  SK-FUNC-INVERSE-HLT · first introduced in Math 31
- R Test whether a function is one-to-one on a domain.
  SK-FUNC-ONE-TO-ONE · first introduced in Math 31

M49-OPS-006

I can find and verify an inverse function algebraically.

Textbook: 1.7 Inverse Functions

SUPPORTING SKILLS

- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Exchange variables and solve to obtain an inverse formula.
  SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY · first introduced in Math 31

M49-OPS-007

I can interpret the relationship between a function and its inverse from graphs or tables.

Textbook: 1.7 Inverse Functions

SUPPORTING SKILLS

- D Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D Reverse input and output units when interpreting an inverse.
  SK-FUNC-INVERSE-UNITS · first introduced in Math 31

Power Functions

REQUIRED

Textbook coverage: 3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation; 3.8 Inverses and Radical Functions

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M49-PWR-001

I can identify a power function and interpret the parameters in y = kx^p.

Textbook: 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

- D Interpret the coefficient and exponent in a power function.
  SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- R Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
  SK-POWER-VOCABULARY · first introduced in Math 39
- D Represent and interpret a power-variation relationship.
  SK-VAR-POWER · first introduced in Math 39

M49-PWR-002

I can construct a power function from two data points or a proportional relationship.

Textbook: 3.9 Modeling Using Variation

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Determine a power function from two points or a proportional relationship.
  SK-POWER-CONSTRUCT · first introduced in Math 39

M49-PWR-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

Textbook: 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

- D Determine power-function behavior as the input approaches zero or positive or negative infinity.
  SK-POWER-END-BEHAVIOR · first introduced in Math 39
- D Relate the coefficient and exponent of a power function to its graph shape.
  SK-POWER-SHAPE · first introduced in Math 39
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-PWR-004

I can recognize and model direct, inverse, and other power-variation relationships.

Textbook: 3.9 Modeling Using Variation

SUPPORTING SKILLS

- D Represent and interpret a direct-variation relationship.
  SK-VAR-DIRECT · first introduced in Math 39
- D Represent and interpret an inverse-variation relationship.
  SK-VAR-INVERSE · first introduced in Math 39
- D Represent and interpret a power-variation relationship.
  SK-VAR-POWER · first introduced in Math 39

M49-PWR-005

I can solve equations involving integer or rational powers.

Textbook: 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21

M49-PWR-007

I can build or select a power-function model for a context and interpret its parameters.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation

SUPPORTING SKILLS

- I Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Determine a power function from two points or a proportional relationship.
  SK-POWER-CONSTRUCT · first introduced in Math 39

M49-PWR-008

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21
- A Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- I Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent.
  SK-POWER-FRACTIONAL-ANALYZE

Polynomial Functions

REQUIRED

Textbook coverage: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

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M49-ALG-002

I can predict the end behavior of a polynomial from its degree and leading coefficient.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I — Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR
- R — Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M49-ALG-003

I can determine polynomial zeros and their multiplicities.

Textbook: 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- I — Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY
- A — Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12
- D — Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-004

I can construct a polynomial function from specified zeros or graphical features.

Textbook: 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D — Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT · first introduced in Math 31
- A — Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A — Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-005

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

Textbook: 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D — Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D — Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A — Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M49-POL-002

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

Textbook: 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I — Use inflection points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-INFLECTION
- I — Use turning points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-TURNING
- I — Use zeros and their multiplicities to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-ZEROS
- A — Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39

M49-POL-001

I can build or select a polynomial model for a context and interpret its features.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D — Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D — Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A — Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A — Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT · first introduced in Math 31

Rational Functions and Limits

REQUIRED

Textbook coverage: 3.7 Rational Functions; 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.3 The Limit Laws; 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 4.4 Graphs of Logarithmic Functions; Chapter 2 Limits; 3.9 Modeling Using Variation

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M49-ALG-006

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

Textbook: 3.7 Rational Functions

SUPPORTING SKILLS

- A — Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- I — Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR
- A — Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21
- I — Recall and correctly use rational-function terms including restriction, hole, asymptote, and end behavior.
  SK-RAT-FUNCTION-VOCAB
- I — Identify a removable discontinuity and its coordinates.
  SK-RAT-HOLE
- I — Determine vertical asymptotes from noncanceling denominator zeros.
  SK-RAT-VERTICAL-ASYMPTOTE
- R — Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
  SK-RAT-VOCABULARY · first introduced in Math 21

M49-ALG-007

I can sketch and analyze a rational function using its algebraic and asymptotic features.

Textbook: 3.7 Rational Functions

SUPPORTING SKILLS

- D — Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR · first introduced in Math 49
- D — Identify a removable discontinuity and its coordinates.
  SK-RAT-HOLE · first introduced in Math 49
- A — Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21
- I — Determine the sign of a rational function on intervals.
  SK-RAT-SIGN-INTERVAL
- D — Determine vertical asymptotes from noncanceling denominator zeros.
  SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-001

I can interpret limit notation as a statement about function behavior near an input.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I — Read and write limit notation with correct approach direction.
  SK-LIM-NOTATION

M49-LIM-002

I can estimate a limit from a graph or table.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I Estimate one-sided and two-sided limits from a graph.
  SK-LIM-GRAPH
- I Estimate a limit from input-output values approaching from both sides.
  SK-LIM-TABLE

M49-LIM-003

I can evaluate a limit algebraically using direct substitution or simplification.

Textbook: 2.1 A Preview of Calculus; 2.3 The Limit Laws

SUPPORTING SKILLS

- I Resolve an indeterminate algebraic form by factoring or rationalizing.
  SK-LIM-SIMPLIFY
- I Test a limit using direct substitution.
  SK-LIM-SUBSTITUTE
- A Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21

M49-LIM-005

I can interpret an infinite limit as vertical-asymptotic behavior.

Textbook: 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I Interpret unbounded output near a finite input as an infinite limit.
  SK-LIM-INFINITE
- D Determine vertical asymptotes from noncanceling denominator zeros.
  SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-006

I can use limit notation to describe behavior at infinity.

Textbook: 3.7 Rational Functions; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I Compare dominant terms to determine behavior at infinity.
  SK-LIM-INFINITY
- D Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR · first introduced in Math 49

M49-LIM-007

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.7 Rational Functions; 4.4 Graphs of Logarithmic Functions; 2.1 A Preview of Calculus; Chapter 2 Limits

SUPPORTING SKILLS

- A Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- D Compare the long-run growth or decay of functions from different families.
  SK-FUNC-DOMINANCE · first introduced in Math 39
- I Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions.
  SK-LIM-GROWTH-ORDER
- A Compare dominant terms to determine behavior at infinity.
  SK-LIM-INFINITY · first introduced in Math 49

M49-ALG-008

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Textbook: 3.7 Rational Functions; 3.9 Modeling Using Variation

SUPPORTING SKILLS

- D Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR · first introduced in Math 49

Sequences and Series

REQUIRED

Textbook coverage: 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

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M49-SEQ-001

I can evaluate and graph a sequence defined explicitly or recursively.

Textbook: 11.1 Sequences and Their Notations

SUPPORTING SKILLS

- I Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT
- I Graph a sequence as discrete ordered pairs.
  SK-SEQ-GRAPH
- I Interpret subscripts and evaluate terms of a sequence.
  SK-SEQ-NOTATION
- I Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE
- I Recall and correctly use the terms sequence, term, index, explicit formula, recursive formula, arithmetic sequence, and geometric sequence.
  SK-SEQ-VOCABULARY

M49-SEQ-002

I can write explicit and recursive formulas for arithmetic sequences.

Textbook: 11.2 Arithmetic Sequences

SUPPORTING SKILLS

- I Recognize constant difference in an arithmetic sequence.
  SK-SEQ-DIFFERENCE
- D Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT · first introduced in Math 49
- D Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-003

I can write explicit and recursive formulas for geometric sequences.

Textbook: 11.3 Geometric Sequences

SUPPORTING SKILLS

- D Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT · first introduced in Math 49
- I Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO
- D Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-004

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

Textbook: 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS

- A Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- D Recognize constant difference in an arithmetic sequence.
  SK-SEQ-DIFFERENCE · first introduced in Math 49
- D Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO · first introduced in Math 49

M49-SEQ-005

I can create and interpret a sequence model for discrete change.

Textbook: 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT · first introduced in Math 49
- I Identify initial value and pattern of change in a discrete context.
  SK-SEQ-MODEL
- A Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SER-001

I can interpret and use summation notation.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Interpret subscripts and evaluate terms of a sequence.
  SK-SEQ-NOTATION · first introduced in Math 49
- I Expand and evaluate a finite sum written in sigma notation.
  SK-SER-SIGMA
- I Recall and correctly use the terms series, partial sum, sigma notation, finite series, infinite series, and convergence.
  SK-SER-VOCABULARY

M49-SER-002

I can calculate a finite arithmetic series.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Recognize constant difference in an arithmetic sequence.
  SK-SEQ-DIFFERENCE · first introduced in Math 49
- I Apply an arithmetic-series sum formula.
  SK-SER-ARITHMETIC

M49-SER-003

I can calculate a finite geometric series.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO · first introduced in Math 49
- I Apply a finite geometric-series sum formula.
  SK-SER-GEOMETRIC-FINITE

M49-SER-004

I can determine whether an infinite geometric series converges and find its sum when it does.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO · first introduced in Math 49
- I Test convergence and sum an infinite geometric series.
  SK-SER-GEOMETRIC-INFINITE

M49-SER-005

I can create and interpret a series model for accumulated change.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Apply an arithmetic-series sum formula.
  SK-SER-ARITHMETIC · first introduced in Math 49
- A Apply a finite geometric-series sum formula.
  SK-SER-GEOMETRIC-FINITE · first introduced in Math 49
- A Test convergence and sum an infinite geometric series.
  SK-SER-GEOMETRIC-INFINITE · first introduced in Math 49
- I Identify the initial term, change factor, and number of terms in an application.
  SK-SER-MODEL

Additional Limit Concepts

EXTENSION

Textbook coverage: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.4 Continuity

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M49-LIM-004

I can determine one-sided limits and decide whether a two-sided limit exists.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- D Estimate one-sided and two-sided limits from a graph.
  SK-LIM-GRAPH · first introduced in Math 49
- D Read and write limit notation with correct approach direction.
  SK-LIM-NOTATION · first introduced in Math 49
- A Estimate a limit from input-output values approaching from both sides.
  SK-LIM-TABLE · first introduced in Math 49

M49-LIM-008

I can distinguish a function value from a limit and determine continuity at a point.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.4 Continuity

SUPPORTING SKILLS

- I Check the three conditions for continuity at a point.
  SK-LIM-CONTINUITY
- A Estimate one-sided and two-sided limits from a graph.
  SK-LIM-GRAPH · first introduced in Math 49
- D Test a limit using direct substitution.
  SK-LIM-SUBSTITUTE · first introduced in Math 49

Parametric Equations

EXTENSION

Textbook coverage: 8.7 Parametric Equations: Graphs; 8.6 Parametric Equations

M49-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

Textbook: 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- D Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

Textbook: 8.6 Parametric Equations

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Eliminate a parameter algebraically.
  SK-PAR-ELIMINATE · first introduced in Math 32
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M49-PAR-003

I can create or interpret parametric equations for motion in the plane.

Textbook: 8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

- D Interpret or construct parametric equations for planar motion.
  SK-PAR-MODEL · first introduced in Math 32
- A Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Conic Sections

EXTENSION

Textbook coverage: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates; 10.3 The Parabola

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M49-CON-001

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- I — Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
  SK-CONIC-CLASSIFY
- I — Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
  SK-CONIC-DISTANCE-DEFINITION
- I — Recall the defining terms and standard notation for circles, parabolas, ellipses, and hyperbolas.
  SK-CONIC-VOCABULARY

M49-CON-002

I can use completing the square to rewrite a general-form conic equation in standard form.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I — Complete squares to convert a general conic equation to standard form.
  SK-CONIC-GENERAL-STANDARD
- D — Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21

M49-CON-003

I can write, interpret, and graph implicit and parametric equations of a circle.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I — Connect implicit, standard, and parametric equations of a circle.
  SK-CONIC-CIRCLE-EQUATIONS
- I — Determine and use a circle's center and radius.
  SK-CONIC-CIRCLE-FEATURES
- A — Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- A — Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-CON-004

I can write, interpret, and graph implicit and parametric equations of an ellipse.

Textbook: 10.1 The Ellipse

SUPPORTING SKILLS

- I Connect implicit, standard, and parametric equations of an ellipse.
  SK-CONIC-ELLIPSE-EQUATIONS
- I Determine and use an ellipse's center, axes, vertices, and foci.
  SK-CONIC-ELLIPSE-FEATURES
- A Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- A Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-CON-005

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

Textbook: 10.2 The Hyperbola

SUPPORTING SKILLS

- I Connect implicit, standard, and parametric equations of a hyperbola.
  SK-CONIC-HYPERBOLA-EQUATIONS
- I Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
  SK-CONIC-HYPERBOLA-FEATURES
- A Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- A Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-CON-006

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- D Determine and use a circle's center and radius.
  SK-CONIC-CIRCLE-FEATURES · first introduced in Math 49
- A Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
  SK-CONIC-CLASSIFY · first introduced in Math 49
- D Determine and use an ellipse's center, axes, vertices, and foci.
  SK-CONIC-ELLIPSE-FEATURES · first introduced in Math 49
- D Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
  SK-CONIC-HYPERBOLA-FEATURES · first introduced in Math 49

M49-CON-007

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- D Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
  SK-CONIC-DISTANCE-DEFINITION · first introduced in Math 49
- A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22

M49-CON-008

I can explain a parabola as the set of points equidistant from a focus and directrix.

Textbook: 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- I Explain a parabola using its focus-directrix distance relationship.
  SK-CONIC-PARABOLA-DEFINITION
- A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22